SGN – Assignment #2

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Exercise 1: Uncertainty propagation

The INTErnational Gamma-Ray Astrophysics Laboratory (INTEGRAL) is a gamma-ray astronomy mission developed by the European Space Agency (ESA) and launched on October 17, 2002. The spacecraft was designed to observe some of the most energetic phenomena in the universe, including gamma-ray bursts, supernova remnants, black holes, and neutron stars.

INTEGRAL has since operated in a highly eccentric orbit around Earth, and after more than two decades of successful operations, INTEGRAL's mission is approaching its end. Due to progressive orbital decay caused by gravitational perturbations and the exhaustion of propellant reserves, ESA has planned a controlled re-entry of the spacecraft into Earth's atmosphere.

As mission controller, you are asked to monitor the orbital evolution to ensure safe and controlled end-of-life disposal in compliance with space debris mitigation guidelines. You have received an estimate of INTEGRAL state at the apocentre epoch $t_0 =$ `2025-10-26T01:10:57.769` (UTC) in terms of its mean and covariance, as reported in Table 1. Assume Keplerian motion can be used to model the spacecraft dynamics.

1. Implement an event function to detect apocentre and pericentre events. Propagate the initial state and report the epoch of the apsis events in the following 5 revolutions.

2. Propagate both initial mean and covariance to all apocenter and pericenter epochs computed at the previous step using both a Linearized Approach (LinCov) and the Unscented Transform (UT). Compare the results in terms of both propagated mean and covariance, considering separately apocenters and pericenters. Elaborate on the results and the differences between the two approaches (UT and LinCov).

3. Perform the same uncertainty propagation process to the same epochs using a Monte Carlo (MC) simulation. Compute the sample mean and sample covariance, and compare them with the results obtained at the previous point. Provide the plot of:

   - The mean state components at the last apocentre and pericentre as a function of the number N of MC samples. Identify the minimum number of samples N_{min} which is required to obtained stable estimates for the state components.
   - the propagated samples of the position of the MC simulation, together with the mean and covariance obtained with all methods, projected on the orbital plane.
   - the time evolution at apocentres and pericentres for all three approaches (MC, LinCov, and UT) of $3\sqrt{\max(\lambda(P_{r,\mathrm{i}}))}$ and $3\sqrt{\max(\lambda(P_{v,\mathrm{i}}))}$, where P_r and P_v are the 3x3 position and velocity covariance submatrices and λ the eigenvalues.

   Compare the results and discuss on the validity of the linear and Gaussian assumption for uncertainty propagation.

Table 1: Estimate of the INTEGRAL state at t_0 provided in ECI J2000.

Parameter	Value
Ref. epoch t_0 [UTC]	2025-10-26T01:10:57.769
Mean state $\hat{\boldsymbol{x}}_0$ [km, km/s]	$\hat{\boldsymbol{r}}_0$ = [7896.37847595, -3152.23575783, 149961.25963557] $\hat{\boldsymbol{v}}_0$ = [-0.6035374932, 0.0159618300, 0.0321154658]
Covariance P_0 [km^2, km^2/s, km^2/s^2]	$\begin{bmatrix} +6.7e-3 & -2.1e-3 & 7.6e-4 & 0 & 0 & 0 \\ -2.1e-3 & +9.7e-3 & +5.3e-4 & 0 & 0 & 0 \\ 7.6e-4 & +5.3e-4 & +2.1e-3 & 0 & 0 & 0 \\ 0 & 0 & 0 & +4.7e-7 & 0 & 0 \\ 0 & 0 & 0 & 0 & +3.1e-7 & 0 \\ 0 & 0 & 0 & 0 & 0 & +1.6e-7 \end{bmatrix}$

1.1 Apsis Event Detection

- **Objective:** To detect pericentre and apocentre epochs of the INTEGRAL spacecraft over five orbital periods using event-driven integration combined with orbital energy-based classification.

- **Methodology & Code Implementation:**

  1. **Orbital Parameter Computation:** Before propagation, the orbital period (T = 63.323 h) and semi-major axis (a = 80656 km) were analytically computed from the initial state $\mathbf{x}_0$ at epoch t_0.
  2. **Event-Driven Propagation:** The spacecraft state was propagated over an interval $[t_0, t_f]$ where $t_f = t_0 + 5.1 \cdot T$, using `ode113` integrator with an integrated event detection mechanism (`odeset` with `'Events', @detect_apo_peri`). The dynamics were modeled using a purely Keplerian two-body formulation.
  3. **Event Function:** The event function monitored the zero-crossings of the radial velocity component:
  $$f(\mathbf{x}) = \mathbf{r} \cdot \mathbf{v} = 0$$
  This condition identifies the apsides, where the spacecraft velocity becomes purely tangential and the geocentric distance reaches a local extremum. The integrator stopped at each zero-crossing, recording both the epoch t_e and the full state vector $\mathbf{x}_e$.
  4. **Apsis Classification:** After event detection, each apsis was classified by comparing its radial distance $r_e = \|\mathbf{r}_e\|$ to the semi-major axis a:
  $$\begin{cases} r_e > a & \Rightarrow \text{Apocentre} \\ r_e < a & \Rightarrow \text{Pericentre} \end{cases}$$
  From the detected events, the first five pericentre passages and the first five apocentre passages were extracted and stored in separate arrays for subsequent uncertainty propagation analysis. The propagation horizon was set to $5.1T$ (rather than exactly $5T$) to ensure all five events of each type were captured.

- **Results & Analysis:**

  The 3D trajectory of the INTEGRAL spacecraft over the 5-orbit arc is visualized in Figure 1. The plot illustrates the high eccentricity of the orbit, with Earth centered at the focus.

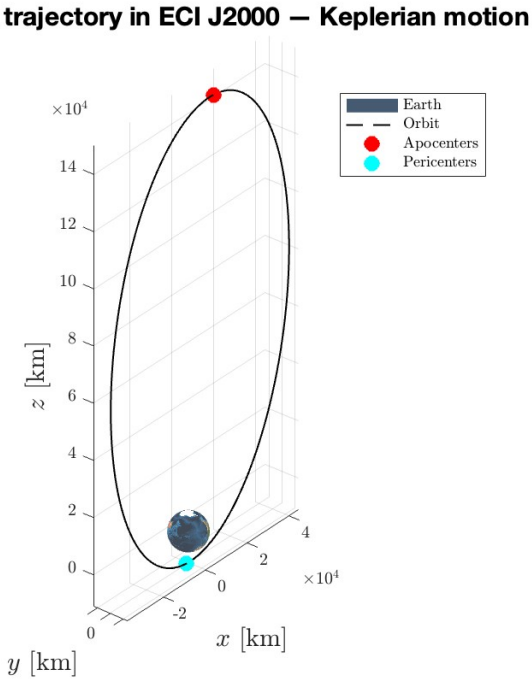


Figure 1: 3D spacecraft trajectory in ECI J2000 frame highlighting the detected apocenters (red) and pericenters (cyan) over the 5-orbit propagation arc.

The numerical state vectors at each detected event are summarized in Table 2 (Pericenters) and Table 3 (Apocenters).

A comparative analysis of the components across the five orbits reveals that the position and velocity values remain constant which confirms the conservative nature of the unperturbed Keplerian model.

Epoch	2025-10-27	2025-10-30	2025-11-01	2025-11-04	2025-11-06
[UTC]	08:50:39	00:10:04	15:29:27	06:48:52	22:08:16
Δt from t_0 [h]	31.662	94.985	158.308	221.631	284.955
r_x [km]	−584.072	−584.072	−584.072	−584.072	−584.072
r_y [km]	233.162	233.162	233.162	233.162	233.162
r_z [km]	−11092.197	−11092.197	−11092.197	−11092.197	−11092.197
v_x [km/s]	8.159542	8.159542	8.159542	8.159542	8.159542
v_y [km/s]	−0.215796	−0.215796	−0.215796	−0.215796	−0.215796
v_z [km/s]	−0.434186	−0.434186	−0.434186	−0.434186	−0.434186

Table 2: Propagated mean state at pericentre passages.

Epoch	2025-10-28	2025-10-31	2025-11-02	2025-11-05	2025-11-08
[UTC]	16:30:22	07:49:46	23:09:10	14:28:34	05:47:58
Δt from t_0 [h]	63.323	126.647	189.970	253.293	316.617
r_x [km]	7896.379	7896.379	7896.379	7896.379	7896.379
r_y [km]	−3152.236	−3152.236	−3152.236	−3152.236	−3152.236
r_z [km]	149961.260	149961.260	149961.260	149961.260	149961.260
v_x [km/s]	−0.603537	−0.603537	−0.603537	−0.603537	−0.603537
v_y [km/s]	0.015962	0.015962	0.015962	0.015962	0.015962
v_z [km/s]	0.032115	0.032115	0.032115	0.032115	0.032115

Table 3: Propagated mean state at apocentre passages.

1.2 Comparative Uncertainty Propagation: LinCov vs. UT

- **Objective:** To propagate the initial estimated mean state $\hat{\mathbf{x}}_0$ and covariance $\mathbf{P}_0$ to the ten detected apsis epochs and evaluate the validity of the linear assumption by comparing the **Linearized Covariance (LinCov)** method against the **Unscented Transform (UT)**.

- **Methodology & Code Implementation:**

  – **Linearized Covariance Propagation (LinCov):** This approach propagates the first two statistical moments (mean and covariance) of the state distribution by linearizing the nonlinear dynamics around the mean trajectory.

  Given an initial mean state $\hat{\mathbf{x}}_0$ at time t_0 and its associated covariance matrix $\mathbf{P}_0$, the mean state $\bar{\mathbf{x}}(t)$ is propagated through the nominal dynamics, and the propagated covariance $\mathbf{P}(t)$ is obtained via the state transition matrix $\mathbf{\Phi}(t, t_0)$, computed by integrating the variational equations:

$$\frac{d\mathbf{\Phi}(\mathbf{x}_0, t_0; t)}{dt} = \mathbf{A}(t)\,\mathbf{\Phi}(\mathbf{x}_0, t_0; t), \quad \text{with} \quad \mathbf{\Phi}(t_0, t_0) = \mathbf{I}_{6\times 6}$$

$$\mathbf{P}(t) = \mathbf{\Phi}(\mathbf{x}_0, t_0; t)\,\mathbf{P}_0\,\mathbf{\Phi}(\mathbf{x}_0, t_0; t)^T$$

  where $\mathbf{A}(t) = \partial\mathbf{f}/\partial\mathbf{x}|_{\mathbf{x}(t)}$ is the Jacobian of the Keplerian vector field evaluated along the nominal trajectory. The integration of the variational equation requires only one integration of the augmented system, eliminating the repeated trajectory propagations that finite-difference column-by-column approximations would demand and avoiding the associated numerical degradation.

  LinCov relies on the assumption that the state uncertainty can be adequately described by a Gaussian distribution for both the initial and propagated distributions.

  – **Unscented Transform (UT):** A derivative-free method for propagating uncertainty through nonlinear systems. Instead of linearizing the dynamics, it approximates the distribution by deterministically sampling a set of $2n + 1$ sigma points around the current mean. These points are propagated through the nonlinear function f, and the statistics are reconstructed from the transformed sigma points. For this analysis, the UT uses $2n + 1$ sigma points wiht $n = 6$ and tuning parameters $\alpha = 10^{-2}$, $\beta = 2$, and $\kappa = 0$. Each sigma point is individually propagated, and the mean and covariance at time t are computed as weighted sums:

$$\bar{\mathbf{y}}(t) = \sum_i W_i^{(m)} \mathbf{Y}_i(t)$$

$$\mathbf{P}(t) = \sum_i W_i^{(c)} \left[\mathbf{Y}_i(t) - \bar{\mathbf{y}}(t)\right] \left[\mathbf{Y}_i(t) - \bar{\mathbf{y}}(t)\right]^T$$

Unlike LinCov, the UT only requires the Gaussian assumption for the initial distribution, allowing for a more accurate capture of non-Gaussian features and the displacement of the true mean (mean shift) as the uncertainty evolves. This is particularly critical for capturing the rotation, skew, and strong state-coupling effects present in highly eccentric orbits.

- **Results & Analysis:** The quantitative comparison of the propagated mean states between LinCov and UT,summarised in Table 4, reveals a divergence between the two methods, particularly at the later epochs.
  - **Pericentre passages:** represent the most nonlinear regime due to the high orbital velocity and steep gravitational gradient near perigee. As shown in Table 4, the position mean difference $\|\bar{\mathbf{x}}_{\mathrm{LC}} - \bar{\mathbf{x}}_{\mathrm{UT}}\|_r$ grows monotonically from 2.25 km at Peri 1 to 107.18 km at Peri 5, while the velocity discrepancy rises from 1.71×10^{-3} km/s to 7.89×10^{-2} km/s. This monotonic growth confirms that the LinCov assumption (that the distribution mean follows the nominal reference trajectory) becomes increasingly inaccurate in high-dynamic regions.
  - **Apocentre passages:** remain in a nearly linear regime. The difference stays below 1 km across all five apocentres (from 0.10 km at Apo 1 to 0.97 km at Apo 5), with velocity discrepancies of order 10^{-5} km/s, confirming that the STM mapping remains adequate where the gravitational potential varies slowly.

Table 4: LinCov vs. UT propagated mean differences at each apsis passage.

Event	$\|\bar{\mathbf{x}}_{\mathrm{LC}} - \bar{\mathbf{x}}_{\mathrm{UT}}\|_r$ [km]	$\|\bar{\mathbf{x}}_{\mathrm{LC}} - \bar{\mathbf{x}}_{\mathrm{UT}}\|_v$ [km/s]
Peri 1	2.2490	1.71×10^{-3}
Peri 2	12.5645	9.32×10^{-3}
Peri 3	33.5440	2.47×10^{-2}
Peri 4	65.0784	4.79×10^{-2}
Peri 5	107.1812	7.89×10^{-2}
Apo 1	0.0985	1.62×10^{-6}
Apo 2	0.2300	4.33×10^{-6}
Apo 3	0.5083	1.03×10^{-5}
Apo 4	0.7143	1.26×10^{-5}
Apo 5	0.9725	1.40×10^{-5}

1.3 Monte Carlo Simulation and Validation

- **Objective:** To establish a high-fidelity statistical benchmark for validating the LinCov and UT methods and to determine the minimum ensemble size $N_{\min}$ required for convergence of the propagated state mean.
- **Methodology & Code Implementation:**
  - **Ensemble Generation:** A Monte Carlo ensemble of $N_{\mathrm{MC}} = 2500$ realizations was generated by sampling the initial distribution $\mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0)$ using `mvnrnd`, with a fixed seed for reproducibility.
  - **Trajectory Propagation:** Each sampled initial state $\mathbf{x}_0^{(i)} \sim \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0)$ was independently propagated through the nonlinear Keplerian equations of motion using the same numerical integration scheme (`ode113` with same settings) employed for the nominal trajectory. The propagated ensemble was then evaluated at the ten

apsis epochs (t_e) previously identified from the nominal solution, yielding clouds of position and velocity realizations at each apocentre and pericentre passage.

- **Statistical Estimation:** At each apsis epoch, the empirical sample mean $\bar{\mathbf{x}}_{\mathrm{MC}}$ and sample covariance $\mathbf{P}_{\mathrm{MC}}$ were computed from the propagated ensemble to serve as the Monte Carlo reference statistics for comparison with LinCov and UT.
- **Convergence Analysis:** The convergence of the sample mean was assessed by computing the running average $\bar{\mathbf{x}}_N$ for increasing ensemble sizes $N = 1, \ldots, N_{\mathrm{MC}}$ and evaluating the component-wise relative error against the full-ensemble reference:

$$\varepsilon_i(N) = \frac{|\bar{x}_i(N) - \bar{x}_i(N_{\mathrm{MC}})|}{|\bar{x}_i(N_{\mathrm{MC}})| + \epsilon} < \mathrm{tol}_{\mathrm{rel}} = 10^{-3}$$

N_{min} is defined as the smallest N for which all six state components simultaneously satisfy the tolerance at both the last apocentre and pericentre.

- **Results & Analysis:** The convergence of the sample mean at the last apocentre and pericentre is illustrated in Figure 2. For small ensemble sizes ($N < 200$), significant stochastic fluctuations are visible, particularly in the velocity components and in the pericentre position, reflecting the larger sample-to-sample variability induced by the nonlinear dynamics near perigee.

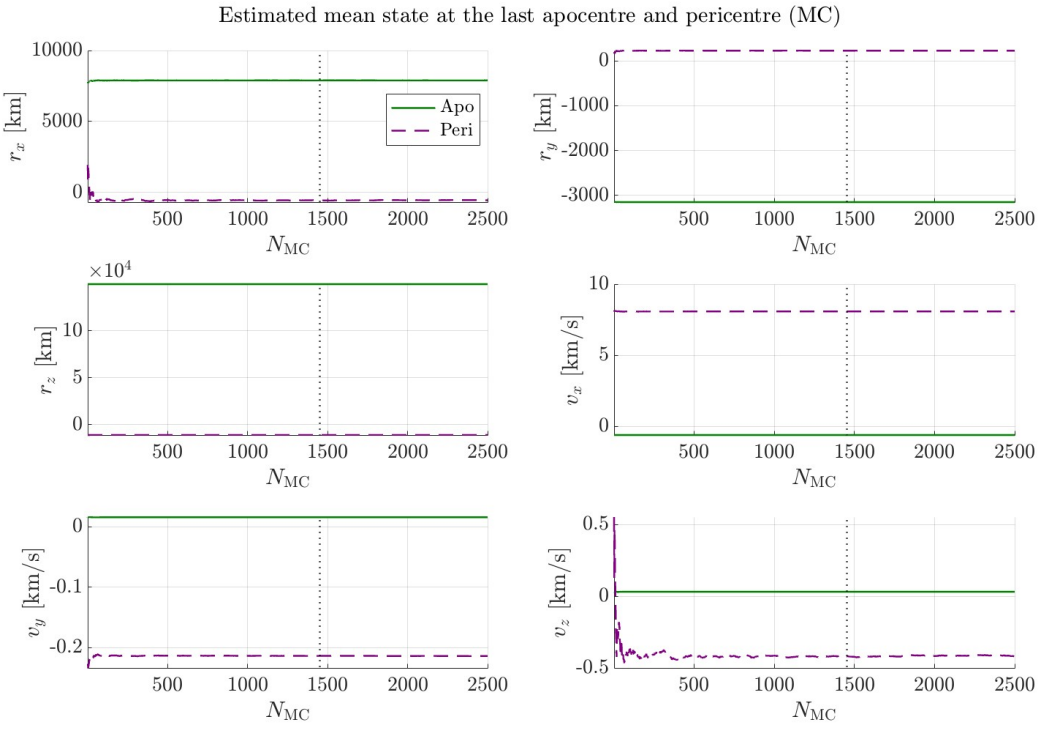


Figure 2: Running Monte Carlo mean for all position and velocity components at the final apocentre (solid green) and pericentre (dashed purple). The vertical dotted line marks the global N_{min} threshold.

The results reveal a clear dependence of convergence speed on orbital phase:

- **Apocentre:** The low orbital velocity and reduced dynamic sensitivity at high altitude lead to rapid convergence, with $N_{\mathrm{min,\,apo}} = 188$.

– **Pericentre:** The increased dynamic sensitivity and stronger nonlinear effects near perigee demand a larger sample size to achieve stable mean estimates, resulting in $N_{\text{min, peri}} = 1453$.

The convergence parameters are summarised in Table 5. The global threshold is $N_{\text{min},Mc} = \max(188\,, 1453) = 1453$, and the adopted ensemble size $N_{\text{MC}} = 2500$ lies well above this value, ensuring that the Monte Carlo statistics provide a reliable reference for evaluating LinCov and UT.

Table 5: Monte Carlo convergence parameters and minimum sample thresholds.

Parameter	**Value**
Relative tolerance tol_{rel}	10^{-3}
$N_{\text{min, apo}}$	188
$N_{\text{min, peri}}$	1453
$N_{\text{min, MC}} = \max(N_{\text{min, apo}},\, N_{\text{min, peri}})$	**1453**

1.4 Final Uncertainty Distribution on the Orbital Plane

- **Objective:** To analyse the geometric structure, orientation, and mean displacement of the propagated uncertainty by projecting the results of all three methods onto the perifocal plane (x_p, y_p) at each apsis passage.
- **Methodology & Code Implementation:**
  - **Perifocal Frame Construction:** For each apsis epoch, the perifocal (orbital plane) reference frame was constructed from the eccentricity vector $\mathbf{e}$ and angular momentum vector $\mathbf{h}$ of the nominal orbit. The transformation matrix $\mathbf{R}_{pf}$ was then assembled to rotate ECI coordinates into the 2D perifocal plane (x_p, y_p).
  - **Covariance Projection:** For LinCov and UT, the 6×6 state covariance matrices were projected onto the perifocal plane by extracting the 3×3 position submatrix $\mathbf{P}_r$ and applying the rotation:
    $$\mathbf{P}_{pf} = \mathbf{R}_{pf}\,\mathbf{P}_r\,\mathbf{R}_{pf}^{\mathsf{T}}$$
    The resulting 2×2 projected covariance was eigendecomposed to construct 1σ uncertainty ellipses, with semi-axes aligned along the principal directions of uncertainty.
  - **Monte Carlo Visualization:** The $N_{\text{MC}} = 2500$ position realizations were projected onto the same perifocal plane and overlaid as a scatter plot to reveal the true distribution geometry. The propagated means from all three methods (LinCov, UT, MC) were also projected and compared to quantify the nonlinear mean displacement induced by the Keplerian dynamics.
- **Results & Analysis:** The projected uncertainty distributions at pericentre and apocentre passages are shown in Figures 3 and 4 respectively.

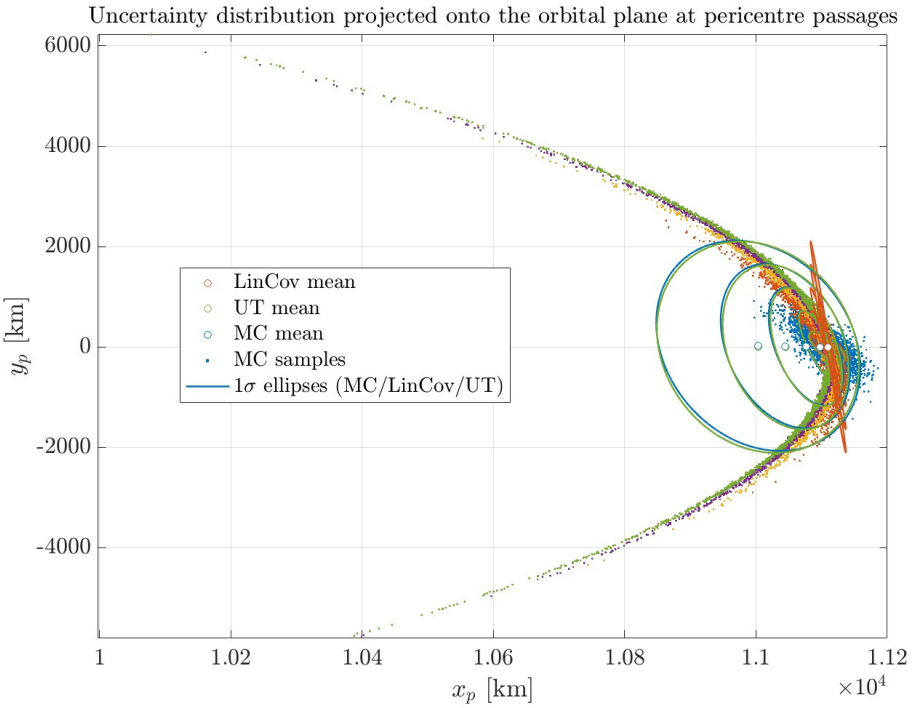


Figure 3: Uncertainty distribution projected onto the perifocal plane at pericentre passages. The MC samples, 1σ ellipses, and propagated means of LinCov, UT, and MC are shown.

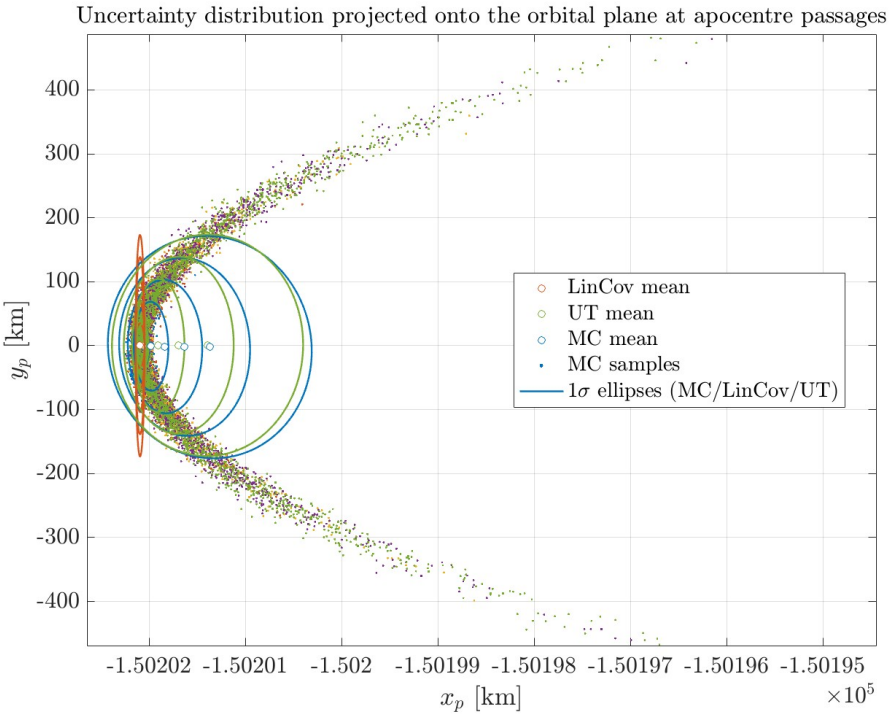


Figure 4: Uncertainty distribution projected onto the perifocal plane at apocentre passages.

– **Nonlinear deformation at pericentre.** As shown in Figure 3, the MC samples form a highly curved, banana-shaped cloud (a classic signature of nonlinear mapping of Gaussian distributions through Keplerian dynamics). This deformation occurs

during high-velocity pericentre passages where the initially Gaussian distribution undergoes significant stretching and rotation along the orbital arc.

  * The **LinCov mean** (red circle) remains fixed at the nominal trajectory point and its 1σ ellipses are narrow, failing to capture the curvature or the true spread of the sample cloud.
  * The **UT mean** (green circle) correctly shifts to track the MC mean (blue diamond), and its ellipses are better aligned with the dominant orientation of the true distribution, demonstrating the superior ability of the UT to account for the nonlinear mean displacement.

– **Near-linearity at apocentre.** At apocentre (Figure 4), the uncertainty magnitude is significantly smaller and the distribution remains approximately symmetric. The LinCov, UT, and MC means lie in close proximity, confirming that the linear assumption remains adequate in the low-dynamic region of the orbit, consistently with the quantitative results reported in Table 4.

1.5 Time Evolution of 3σ Uncertainty

- **Objective:** To evaluate the temporal growth of the uncertainty magnitude at each apsis epoch using a direction-independent scalar metric, and to compare the predictions of LinCov, UT, and Monte Carlo.

- **Methodology & Code Implementation:**

  – **Uncertainty Metric Definition:** To evaluate the temporal growth of uncertainty magnitude in a direction-independent manner, a scalar 3σ metric was adopted based on the maximum eigenvalue of the covariance submatrices:

$$\sigma_{3,r} = 3\sqrt{\max\big(\lambda_i(\mathbf{P}_r)\big)}, \qquad \sigma_{3,v} = 3\sqrt{\max\big(\lambda_i(\mathbf{P}_v)\big)}$$

  where $\mathbf{P}_r$ and $\mathbf{P}_v$ are the 3×3 position and velocity covariance submatrices, and λ_i denote their eigenvalues. This metric corresponds to the largest semi-axis of the 3σ confidence ellipsoid, providing a worst-case bound on uncertainty.

  – **Evaluation at Apsis Epochs:** For each of the three methods (LinCov, UT, MC), the covariance matrices at all ten apsis epochs (five apocentres and five pericentres) were post-processed. At each epoch, the 6×6 covariance was partitioned into position and velocity blocks, eigenvalues were computed, and the maximum eigenvalue was extracted to construct the scalar bounds $\sigma_{3,r}$ and $\sigma_{3,v}$.

  – **Temporal Evolution Comparison:** The computed 3σ bounds were plotted as time series to visualize uncertainty growth across successive orbital revolutions and to compare the predictions of the three propagation methods.

- **Results & Analysis:** The time evolution of the maximum 3σ bounds is shown in Figure 5.

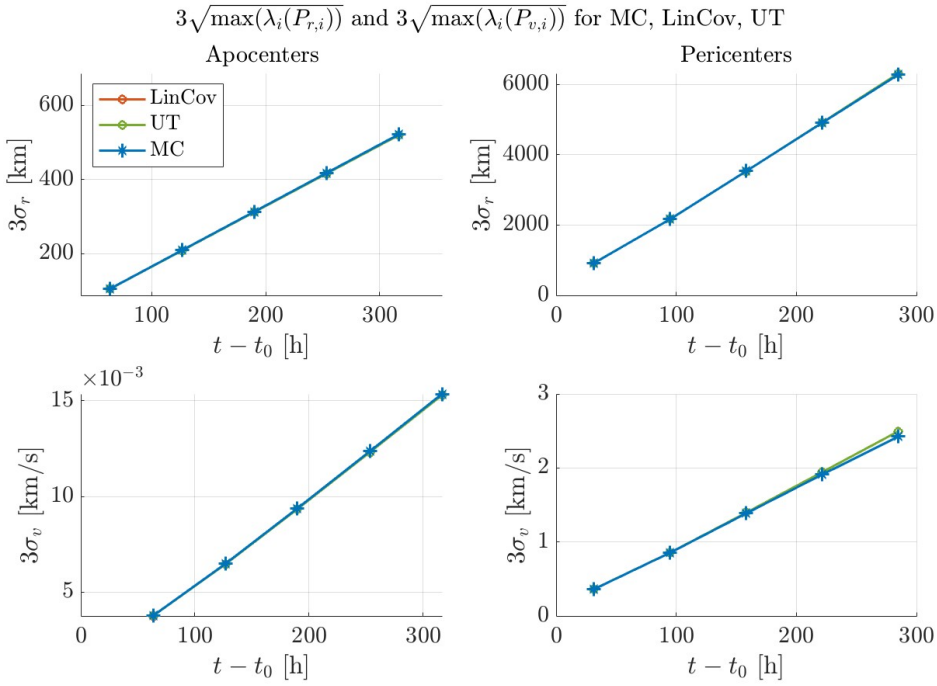


Figure 5: Time evolution of $3\sqrt{\max(\lambda_i(P_{r,i}))}$ and $3\sqrt{\max(\lambda_i(P_{v,i}))}$ at apocentre (left) and pericentre (right) for LinCov, UT, and MC.

- **Monotonic growth:** 3σ bounds grow monotonically for all methods, reflecting uncertainty accumulation under Keplerian dynamics over successive orbital revolutions.
- **Pericentre vs. apocentre:** Uncertainty is larger at pericentre (∼6000 km, ∼2.5 km/s at Peri 5) than apocentre (∼500 km, ∼0.015 km/s at Apo 5), due to the stronger gravitational gradient and higher velocity near perigee which is expected from the anlysis performed in the previous section.
- **Method agreement:** LinCov, UT, and MC yield nearly identical 3σ values at every epoch; however, this does not imply equivalent uncertainty distributions as orbital-plane projections shown before (Figures 3, 4) reveal non-Gaussian deformation and mean shift at pericentre, captured only by UT and MC.

Exercise 2: Batch filters

You have been asked to track INTEGRAL to improve the accuracy of its state estimate. To this aim, you shall schedule the observations from the two ground stations reported in Table 6.

1. *Compute visibility windows.* You have received an estimate of INTEGRAL state at the apocentre epoch $t_0 =$ `2025-10-26T01:10:57.769` (UTC) in terms of its mean and covariance, as reported in Table 1. Propagate this state assuming Keplerian motion over a uniform time grid (with a time step of 60 seconds) and compute all the visibility time windows from the available stations in the time interval from $t_i =$ `2025-10-27T07:10:57.769` (UTC) to $t_f =$ `2025-10-28T10:00:00.000` (UTC). Report the resulting visibility windows and plot the predicted Azimuth and Elevation profiles within these time intervals.

2. *Simulate measurements.* The Two-Line Elements (TLE) set of INTEGRAL are reported in Table 7 (and the file provided in WeBeep as 27540.3le). Use SGP4 and the provided TLE to simulate the measurements acquired by the sensor network in Table 6 by:

(a) Computing the spacecraft position over the visibility windows identified in Point 1 and deriving the associated expected measurements.

(b) Simulating the measurements by adding a random error to the expected measurements (assume a Gaussian model to generate the random error, with noise provided in Table 6). Discard any measurements (i.e., after applying the noise) that do not fulfill the visibility condition for the considered station.

Verify (graphically) that the applied *noise level* is within the desired boundary through a comparison with the ground-truth trajectory (the one propagated with SGP4).

3. *Solve the navigation problem.* Using the measurements simulated at the previous point:

   (a) Find the least squares (minimum variance) solution to the navigation problem without a priori information using
   - the epoch t_i as estimation epoch;
   - the true state as the state derived from the TLE set in Table 7 at the estimation epoch;
   - only the simulated angular measurements obtained with the TROLL ground station;
   - pure Keplerian motion to model the spacecraft dynamics.

   (b) Repeat step 3a by using all simulated measurements from the TROLL ground station, i.e., angles and range.

   (c) Repeat steps 3a and 3b by using a J2-perturbed motion to model the spacecraft dynamics.

   Provide the results in terms of navigation solution*, together with the square root of the trace of i) the estimated covariance submatrix of the position elements and ii) estimated covariance submatrix of the velocity elements. Finally, considering a linear mapping of the estimated covariance from Cartesian state to Keplerian elements, provide the standard deviation associated to the semimajor axis. Elaborate on the results, comparing the different solutions.

4. *Extended visibility.* To improve the accuracy of the orbit determination process, you leverage the measurements acquired by the KOUROU ground station. For this purpose, repeat the operations performed in Point 3 by including the simulated measurements acquired from KOUROU. Specifically:

   (a) Find the least squares (minimum variance) solution to the navigation problem without a priori information using
   - the epoch t_i as estimation epoch;
   - the true state as the state derived from the TLE set in Table 7 at the estimation epoch;
   - all the simulated measurements (angles and range) obtained with the two ground stations;
   - pure Keplerian motion to model the spacecraft dynamics.

   (b) Repeat step 4a by using a J2-perturbed motion to model the spacecraft dynamics.

   Provide the results in terms of navigation solution, together with the square root of the trace of i) the estimated covariance submatrix of the position elements and ii) estimated covariance submatrix of the velocity elements, and standard deviation associated with

*Not just estimated state or covariance

the semimajor axis. Elaborate on the results, comparing the different solutions with the previous case.

5. *Performance assessment.* To assess the validity of your results, you are asked to compare the estimated trajectories with the ground-truth (the one propagated with SGP4). To do so:

   (a) Compute the ground-truth trajectory through SGP4 for 6 orbital periods of INTEGRAL starting from t_i.

   (b) Compute the six estimated trajectories over the same time interval by using, as initial conditions, the state estimates obtained in the six previous cases. These correspond to: (i) TROLL with angles-only, (ii) TROLL with angles-only including J2, (iii) TROLL with angles+range, (iv) TROLL with angles+range including J2, (v) TROLL+KOUROU with angles+range, and (vi) TROLL+KOUROU with angles+range including J2.

   (c) For each case, plot separately the time evolution of the position error norm and the velocity error norm. Compare the solutions and elaborate on the results.

Table 6: Sensor network to track INTEGRAL: list of stations, including their features.

Station name	KOUROU	TROLL
Coordinates	LAT = 5.25144° LON = -52.80466° ALT = -14.67 m	LAT = -72.011977° LON = 2.536103° ALT = 1298 m
Type	Radar (monostatic)	Radar (monostatic)
Measurements type	Az, El [deg] Range (one-way) [km]	Az, El [deg] Range (one-way) [km]
Measurements noise (diagonal noise matrix R)	$\sigma_{Az,El}$ = 125 mdeg σ_{range} = 0.01 km	$\sigma_{Az,El}$ = 125 mdeg σ_{range} = 0.01 km
Minimum elevation	0 deg	0 deg

Table 7: TLE of INTEGRAL.

```
1 27540U 02048A   25298.18996762  .00000023  00000-0  00000-0 0  9991
2 27540  91.0900 358.3898 8628526 267.2169  62.6507  0.37900391 25750
```

2.1 Compute Visibility Windows

- **Objective:** To determine the time intervals during which INTEGRAL is visible from the KOUROU and TROLL ground stations by propagating the Keplerian mean state over the analysis interval $[t_i,\, t_f]$.

- **Methodology & Code Implementation:**

  1. **Initial State Propagation:** The reference mean state at t_0 (Table 1) was propagated forward from t_0 to the analysis start epoch t_i under pure Keplerian dynamics using `keplerian_STM_propagator`, yielding the initial condition $\mathbf{x}(t_i)$ for the visibility analysis window. For the subsequent orbit determination points, an additional propagation under a J_2 model is also performed to provide a consistent initial guess with respect to the model employed in the batch least squares solution.
  2. **Trajectory Generation:** The state $\mathbf{x}(t_i)$ was then propagated over the full interval $[t_i, t_f]$, with $t_i =$ `2025-10-27T07:10:57.769` UTC and $t_f =$ `2025-10-28T10:00:00` UTC, using a uniform time step of $\Delta t = 60\,\mathrm{s}$, producing a discrete predicted ECI trajectory sampled at epochs $t_k = t_i + k \cdot \Delta t$ for $k = 0, 1, \ldots, N$.
  3. **Topocentric Coordinate Computation:** At each epoch t_k, the spacecraft position was transformed into the topocentric reference frame of each ground station (KOUROU and TROLL) via the following formula:

  $$\mathbf{x}_{\mathrm{rel}}^{\mathrm{TOPO}}(t_k) = \mathbf{R}_{J2000\to\mathrm{TOPO}}(t_k)\, \mathbf{x}_{\mathrm{rel}}^{J2000}(t_k), \tag{1}$$

  yielding the slant range ρ, azimuth Az, and elevation El. The procedure is summarized in Algorithm 6 and implemented through the function `antenna_pointing`. The azimuth is wrapped into $[0, 2\pi)$ for plotting.

Algorithm 1 Antenna Pointing Computation

```
Define the topocentric reference frame based on the ground station
for each time step do
    Retrieve the ground station position in the inertial frame
    Compute the relative position vector between the spacecraft and the station
    Transform this vector into the topocentric frame
    Determine the spacecraft position in the local topocentric frame
    Convert the position to azimuth, elevation, and range
    Store the results
end for
Convert angular values to degrees
return azimuth, elevation, and range profiles
```

Figure 6: Antenna Pointing algorithm.

  4. **Visibility Window Extraction:** A visibility flag is raised whenever $El \geq 0°$ (Table 6); contiguous flagged epochs define each window for each station, stored for subsequent measurement simulation.

- **Results & Analysis:** One visible pass was identified for each station. The computed windows are reported in Table 8. The corresponding azimuth, elevation, and range profiles are shown in Figures 7 and 8, while the azimuth–elevation profiles are shown in Figures 9 and 10.

Table 8: Computed visibility windows for TROLL and KOUROU.

Station	Start [UTC]	End [UTC]
TROLL	2025-10-27T08:00:57.769	2025-10-27T09:09:57.769
KOUROU	2025-10-27T18:43:57.769	2025-10-28T07:30:57.769

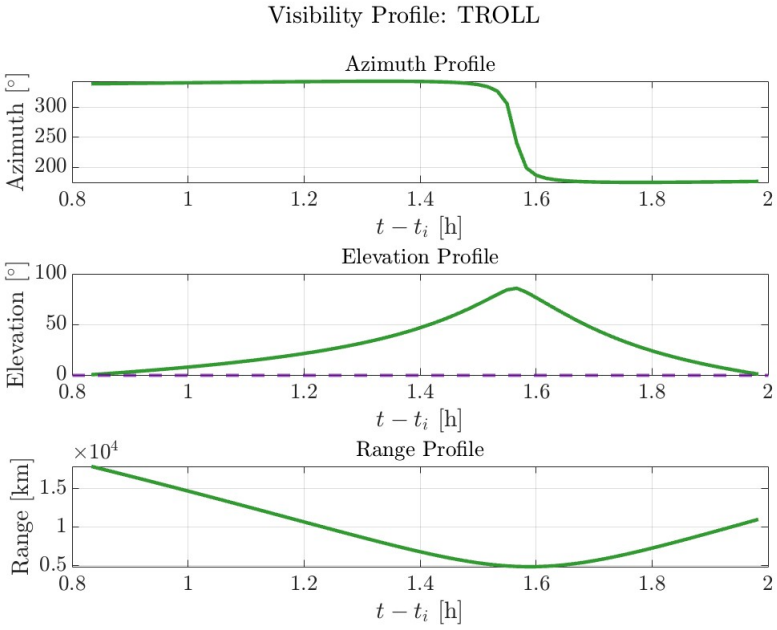


Figure 7: Az, El, and ρ profiles over the TROLL visibility window. Dashed line: $El_{\min} = 0°$.

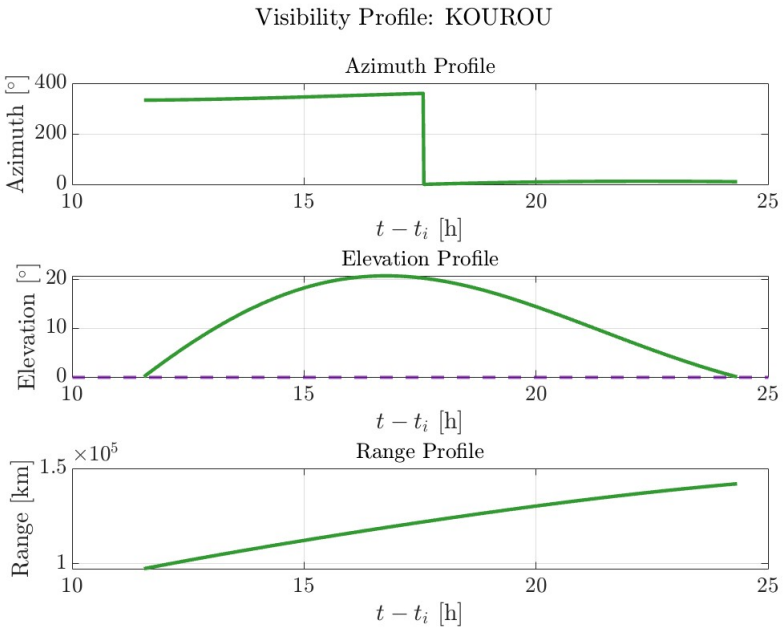


Figure 8: Az, El, and ρ profiles over the KOUROU visibility window. Dashed line: $El_{\min} = 0°$.

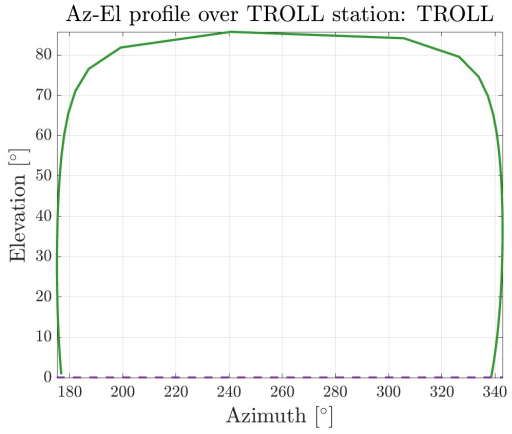


Figure 9: Azimuth–elevation profile for TROLL during its visibility window.

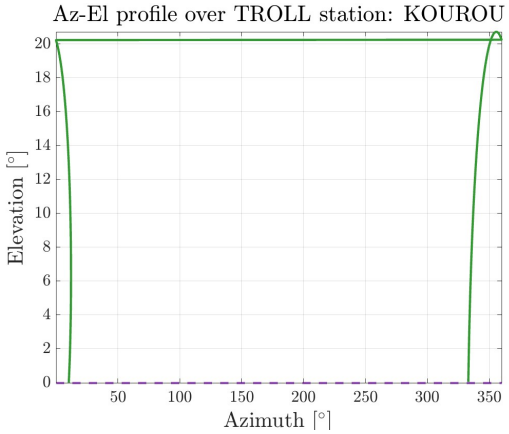


Figure 10: Azimuth–elevation profile for KOUROU during its visibility window.

The two passes reflect the HEO geometry of INTEGRAL:

- **TROLL** (∼69 min): short pass, peak elevation ∼85°, rapid azimuth variation, minimum range ∼5 000 km which is signature of the fast-moving perigee passing nearly overhead.
- **KOUROU** (∼12 h 47 min): long pass, low broad elevation peaking at ∼20°, range $\sim 10^5$ km which is consistent with tracking the slow-moving apogee from an equatorial station, representing a less favourable geometry.

2.2 Simulate Measurements

- **Objective:** To generate a realistic set of noisy angular and range measurements for KOUROU and TROLL using the SGP4 ground-truth trajectory, and to verify graphically that the applied noise level is consistent with the specified sensor accuracy.
- **Methodology & Code Implementation:**

1. **Ground-Truth Trajectory Generation:** The satellite state was propagated using the SGP4 analytical model initialized with the TLE data in Table 7 to generate the ground truth trajectory $x_{true}(t)$ using parameters: `run_type='u'`, `ops_mode='a'` and `wgs_const=72` (WGS-72 gravity model). SGP4 incorporates simplified perturbations including atmospheric drag and non-uniform Earth gravity effects (J_2–J_4). At each epoch t_k within the visibility windows, the TEME-frame output was converted to ECI J2000 via:

$$\mathbf{x}^{J2000}(t_k) = [P(t_k)][N(t_k)]\,[R_z(\Delta\psi\cos\bar{\epsilon})]^T\,\mathbf{x}^{\mathrm{TEME}}(t_k), \tag{2}$$

where $[P(t_k)]$ (precession) is computed from TDT centuries from J2000.0, and $[N(t_k)]$ (nutation) is constructed from corrected offsets $\Delta\psi = -0.118679$ arcsec and $\Delta\epsilon = -0.009495$ arcsec (converted to radians) and the third term accounts for TEME-to-TOD rotation (via SPICE routine `teme2eci`). The resulting trajectory $\mathbf{x}_{\mathrm{true}}(t_k)$ served as the ground-truth reference.

2. **Ideal Measurement Computation:** Noise-free ideal observables $Y_{ideal}(t_k)$ were generated at candidate epochs within the visibility windows (Subsection 2.1) sampled at $\Delta t_{\mathrm{meas}} = 60$ s. For each candidate epoch t_k, the SGP4 ground-truth state $\mathbf{x}_{\mathrm{true}}(t_k)$ was transformed to topocentric observables via the `antenna_pointing_rad` function.

$$\mathbf{Y}_{\text{true}}(t_k) = h(\mathbf{x}_{\text{true}}(t_k)) = \begin{bmatrix} Az_k \\ El_k \\ \rho_k \end{bmatrix}$$

Station-specific candidate counts: TROLL ($N_{\text{cand}} = 70$), KOUROU ($N_{\text{cand}} = 768$).

3. **Noise Injection:** Zero-mean Gaussian noise was added element-wise to each measurement. The measurement noise covariance matrix is diagonal:

$$\mathbf{R} = \text{diag}\big(\sigma_{Az}^2,\, \sigma_{El}^2,\, \sigma_{\rho}^2\big)$$

Independent Gaussian samples $\nu_i \sim \mathcal{N}(0, \sigma_i^2)$ were generated via `randn()` (with fixed seed for reproducibility) and added to produce:

$$\mathbf{Y}_{\text{meas}}(t_k) = \mathbf{Y}_{\text{ideal}}(t_k) + \boldsymbol{\nu}_k$$

4. **Validity Filtering:** After noise injection, any measurement with elevation $El < 0°$ was discarded to ensure physical consistency (satellite below horizon). This post-filtering reduced the TROLL dataset from 70 candidate epochs to 69 valid measurements, and the KOUROU dataset from 768 to 764.

- **Results & Analysis:**
  - Residual time histories (Figures 11–14) show zero-mean scatter around truth with the large majority of samples within $\pm 3\sigma$ bounds, where the residuals are defined as $\Delta Az_k = Az_{\text{meas},k} - Az_{\text{true},k}$, $\Delta El_k = El_{\text{meas},k} - El_{\text{true},k}$, $\Delta\rho_k = \rho_{\text{meas},k} - \rho_{\text{true},k}$, with azimuth residuals remapped via $\text{atan2}(\sin(\Delta Az_k), \cos(\Delta Az_k))$ to avoid 0–2π discontinuities.
  - The empirical standard deviations match the specified noise levels ($\sigma_\rho \approx 0.01$ km, $\sigma_{Az,El} \approx 0.125$), with a small fraction of outliers expected due to finite sample statistics and the elevation mask applied after noise injection. Normalized residual histograms are consistent with the Gaussian noise model: KOUROU shows approximately centered distributions, while TROLL exhibits larger sampling oscillations due to smaller sample size, though residual statistics remain aligned with the specified levels. Both datasets confirm the noise injection is statistically sound and ready for batch least-squares processing in the upcoming points.

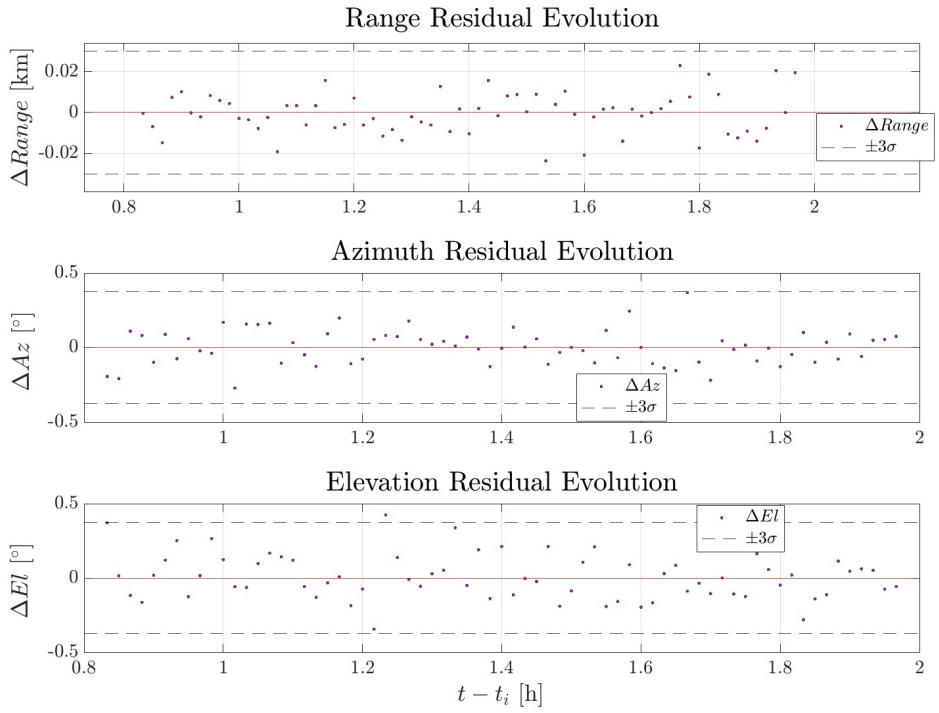


Figure 11: Measurement residual evolution ($\Delta\rho$, ΔAz, ΔEl) for TROLL. .

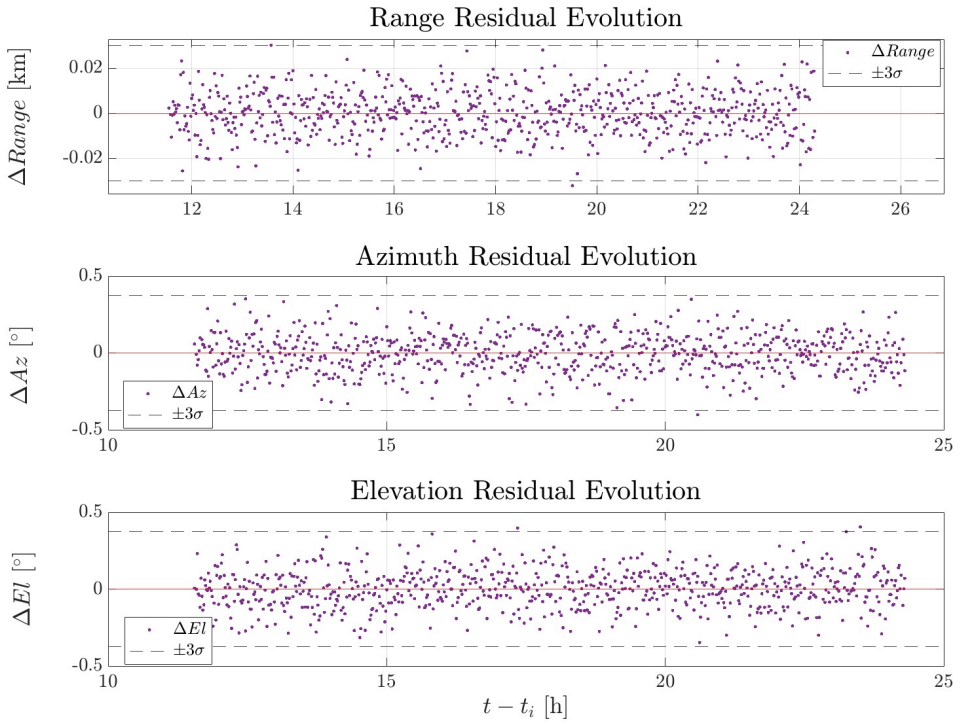


Figure 12: Measurement residual evolution ($\Delta\rho$, ΔAz, ΔEl) for KOUROU. The larger number of measurements (764) provides a denser scatter, uniformly contained within $\pm 3\sigma$.

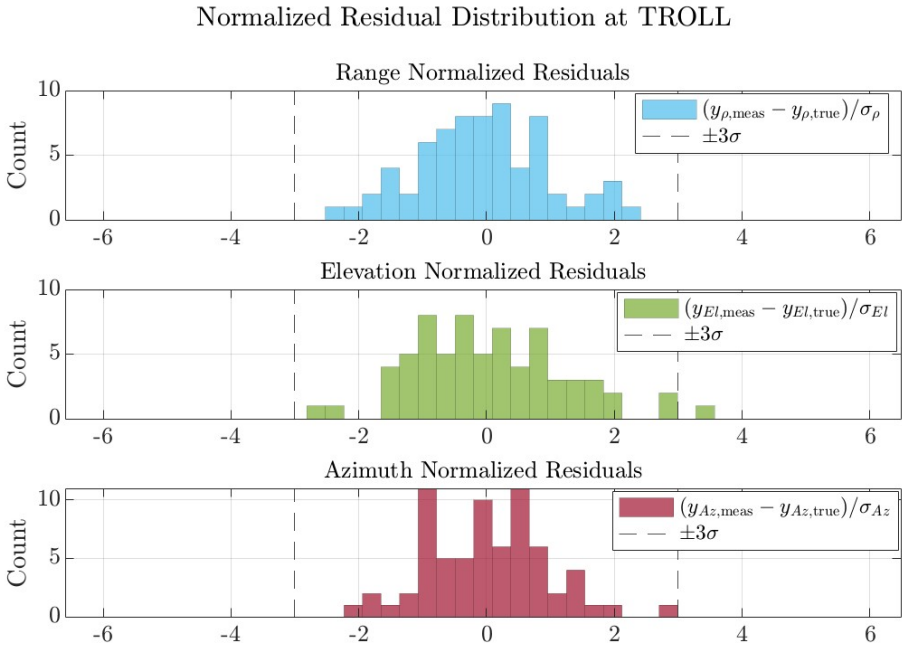


Figure 13: Normalized residual distributions at TROLL for range, elevation, and azimuth.

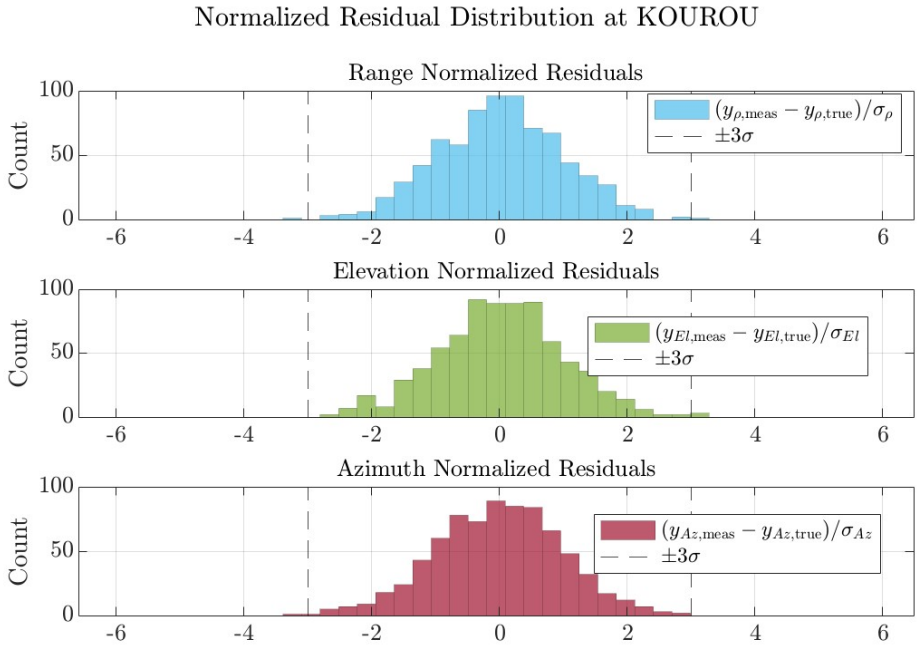


Figure 14: Normalized residual distributions at KOUROU for range, elevation, and azimuth.

2.3 Navigation Problem Solution (Single Station)

- **Objective:** To determine the optimal state estimate $\hat{\mathbf{x}}$ of INTEGRAL at the estimation epoch t_i by minimizing the residuals between simulated measurements and their model-based predictions, and to evaluate the impact of measurement type (angles-only vs. angles+range) and dynamical model fidelity (Keplerian vs. J_2-perturbed) across four single-station cases (i–iv) using TROLL observations. The multi-station cases involving

KOUROU will be treated in the next section.

- **Methodology & Code Implementation:** The navigation problem was solved through a weighted batch least squares estimation framework following a minimum variance approach without *a priori* information, implemented using the Levenberg–Marquardt algorithm via MATLAB's `lsqnonlin` function.

  1. **Problem Formulation:** The optimal state estimate $\hat{\mathbf{x}}$ is found by solving the nonlinear least squares problem:

  $$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \frac{1}{2} \sum_{k=1}^{m} \big(\mathbf{y}_k - h(\mathbf{x}, t_k)\big)^T W_k \big(\mathbf{y}_k - h(\mathbf{x}, t_k)\big) \tag{3}$$

     – $J(\mathbf{x}) = \frac{1}{2}\mathbf{r}^T W \mathbf{r}$ is the weighted cost function, with $\mathbf{r} = \mathbf{Y} - h(\mathbf{x})$ the stacked residual vector collecting all measurement epochs, $\mathbf{Y} = [\mathbf{y}_1, \ldots, \mathbf{y}_m]^T$ the concatenated measurement vector.
     – $h(\mathbf{x}, t_k)$ is the nonlinear observation model that propagates the state $\mathbf{x}$ to each epoch and computes predicted observables via `antenna_pointing_rad` function (note: Azimuth residuals are evaluated through an angular difference wrapped in $[-\pi, \pi]$ using a sine-cosine formulation to avoid discontinuities).
     – $W = R^{-1}$ is the weight matrix constructed from the measurement noise covariance $\mathbf{R} = \mathrm{diag}(\sigma_{Az}^2, \sigma_{El}^2, \sigma_\rho^2)$; and the cost is the weighted sum of squared residuals.

  2. **Initial Guess Generation:** An initial guess for the state at the estimation epoch t_i was obtained by propagating the TLE mean elements (Table 7) from the TLE reference epoch to t_i using the SGP4 analytical propagator. This provided a reasonable starting point $\mathbf{x}_0^{\text{guess}}$ for the iterative optimization.

  3. **Dynamical Model Selection:** The measurement prediction function $h(\mathbf{x}, t_k)$ was implemented using either pure Keplerian two-body dynamics or Keplerian dynamics augmented with J_2 oblateness effects, depending on the case configuration. For J_2-perturbed cases, the gravitational acceleration was first computed in the Earth-Centered Earth-Fixed (ECEF) frame:

  $$\mathbf{a}_{J_2,\text{ECEF}} = \frac{3}{2}\mu J_2 \left(\frac{R_{\text{EARTH}}}{r}\right)^2 \frac{1}{r^3} \begin{bmatrix} x\big(5(z/r)^2 - 1\big) \\ y\big(5(z/r)^2 - 1\big) \\ z\big(5(z/r)^2 - 3\big) \end{bmatrix} \tag{4}$$

     and then rotated into the ECI J2000 frame using SPICE transformation routines before integration.

  4. **Cost Function Evaluation:** For a given trial state $\mathbf{x}$, the cost function was evaluated by: (a) propagating $\mathbf{x}$ from t_i to each measurement epoch using the selected dynamical model; (b) computing the predicted observables at each epoch via the topocentric transformation; (c) forming the weighted residual vector $\mathbf{r} = W^{1/2}(\mathbf{Y} - h(\mathbf{x}, t_k))$. The optimizer (`lsqnonlin`) iteratively adjusted $\mathbf{x}$ to minimize $\|\mathbf{r}\|^2$.

  5. **Covariance Estimation:** Upon convergence, the *a posteriori* covariance matrix $\mathbf{P}_x$ was computed from the Jacobian H of the measurement function evaluated at the optimal state:

  $$\mathbf{P}_x = \hat{\sigma}^2 \big(H^T W H\big)^{-1} \tag{5}$$

     where the estimated variance of unit weight is $\hat{\sigma}^2 = \frac{2J(\hat{\mathbf{x}})}{m-n}$ (with $J(\hat{\mathbf{x}})$ being the minimized cost function value, m the number of scalar residual components, and $n = 6$ the state parameters), which accounts for potential model mismatch or underestimated measurement noise.

6. **Semimajor Axis Uncertainty Propagation:** The uncertainty in the semimajor axis a (i.e σ_a) was derived via first-order linear covariance mapping. To avoid complex analytical derivatives, the Jacobian $F = \nabla_{\mathbf{x}} a = \frac{\partial a}{\partial \mathbf{x}}$ is computed numerically via central differences by perturbing each state component by a step $h_i = \max(\sqrt{\epsilon}, \sqrt{\epsilon}|x_i|)$ with ϵ representing machine precision:

$$\begin{cases} a = -\dfrac{\mu}{2\varepsilon}, \quad \left(\text{where } \varepsilon = \dfrac{v^2}{2} - \dfrac{\mu}{r}\right) \\ F_i \approx \dfrac{a(\mathbf{x} + h_i \mathbf{e}_i) - a(\mathbf{x} - h_i \mathbf{e}_i)}{2h_i} \end{cases} \implies P_a = F \mathbf{P}_x F^T \implies \sigma_a = \sqrt{P_a} \tag{6}$$

where e_i is the i-th unit basis vector.

This procedure was executed for all six navigation cases (i–vi), varying the measurement type (angles-only vs. angles+range), dynamical model (Keplerian vs. J_2), and station configuration (TROLL-only vs. TROLL+KOUROU).

- **Results & Analysis:** The formal uncertainties are summarised in Table 9 and the estimated state vectors are reported in Table 10.

Table 9: State uncertainties at estimation epoch t_i.

Case	σ_r [km]	σ_v [km/s]	σ_a [km]
(i) TROLL Ang (Kep)	96.859	0.0252	3257.668
(ii) TROLL Ang (J_2)	97.746	0.0253	3251.745
(iii) TROLL All (Kep)	29.713	0.0035	34.295
(iv) TROLL All (J_2)	40.017	0.0047	47.367

Table 10: Navigation solution at t_i using TROLL measurements.

Case	r_x [km]	r_y [km]	r_z [km]	v_x [km/s]	v_y [km/s]	v_z [km/s]
(i)	−27416.79	566.99	12237.09	2.00378	0.02280	−4.20636
(ii)	−27409.87	567.01	12236.41	2.00189	0.02283	−4.20605
(iii)	−27366.95	451.22	12248.52	1.99053	0.02888	−4.20731
(iv)	−27373.64	450.84	12238.83	1.99219	0.02525	−4.20645

The inclusion of range measurements (cases iii and iv) reduces σ_a from ∼3250 km in the angles-only cases to below 50 km, and σ_v from ∼25 m/s to ∼5 m/s. This confirms that angles-only data provides poor observability of the orbital energy, while range strongly constrains the radial distance which reduces the uncertainty evidently. The Keplerian–J_2 model choice has limited effect over the short TROLL window. The residuals for case (i, angles-only) in Figure 15 show large scatter reflecting poor observability, while case (iv, angles+range J_2) in Figure 16 exhibits zero-centred convergence within $\pm 3\sigma$ bounds.

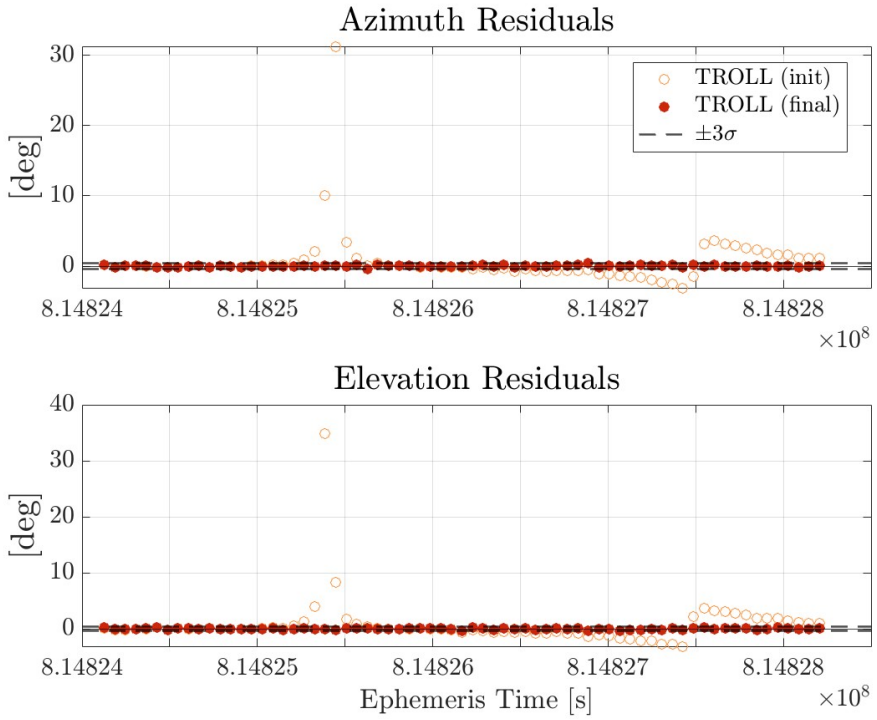


Figure 15: Unweighted measurement residuals for case (i): TROLL angles-only (Keplerian).

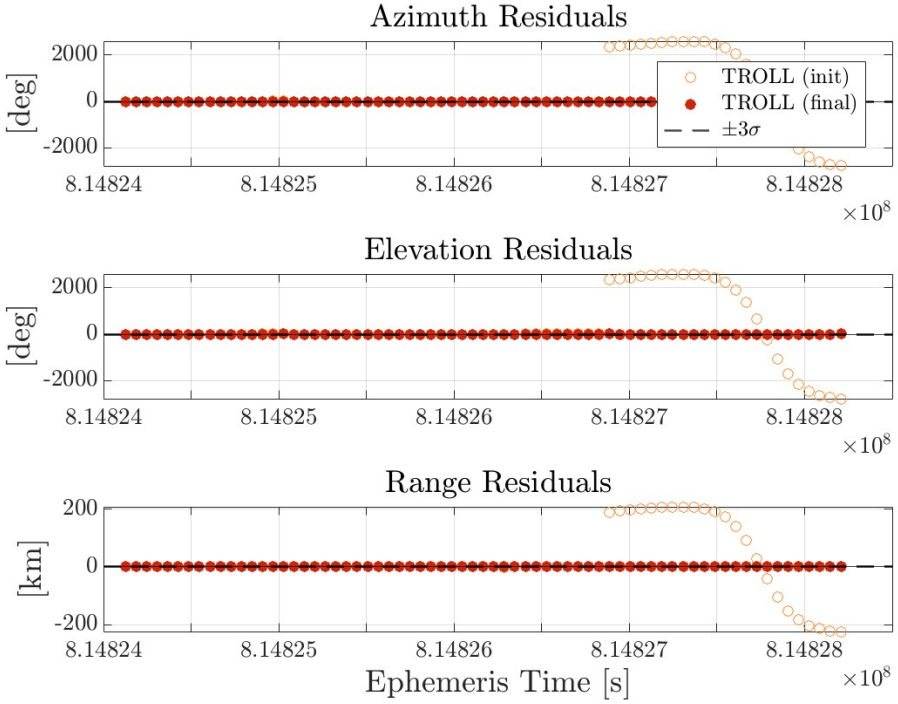


Figure 16: Unweighted measurement residuals for case (iv): TROLL angles+range (J_2).

2.4 Extended Visibility (Multi-Station)

- **Objective:** To improve orbit determination accuracy by leveraging the geometric diversity and extended visibility of a two-station network. Cases (v) and (vi) repeat the batch least-squares estimation using combined measurements (Az, El, ρ) from both TROLL and KOUROU, comparing Keplerian and J_2-perturbed dynamical models.

- **Methodology & Code Implementation:**

  - The multi-station navigation problem extends the single-station framework by fusing observation sets from the TROLL (69 valid samples) and KOUROU (764 valid samples) stations. The estimation remains a weighted batch least-squares problem, where the cost function processes a concatenated residual vector.
  - Each station block is constructed by propagating the estimated state $\hat{\mathbf{x}}$ from t_i to the respective measurement epochs using the selected dynamical model (Keplerian or J_2). Predicted observables are generated via the `antenna_pointing_rad` function. As in the single-station case, azimuth residuals are evaluated through a wrapped angular difference to maintain circular continuity.
  - To account for the measurement noise, each station block is scaled using a block-diagonal weight matrix constructed via the Kronecker product ($\otimes$) before being stacked into the unified global residual vector $\mathbf{r}$ for the optimizer:

$$\mathbf{r} = \begin{bmatrix} \mathbf{r}_{\text{TROLL}} \\ \mathbf{r}_{\text{KOUROU}} \end{bmatrix} = \begin{bmatrix} (\mathbf{I}_{N_{\text{T}}} \otimes W)\big(\mathbf{Y}_{\text{TROLL}} - \mathbf{H}_{\text{TROLL}}(\mathbf{x})\big) \\ (\mathbf{I}_{N_{\text{K}}} \otimes W)\big(\mathbf{Y}_{\text{KOUROU}} - \mathbf{H}_{\text{KOUROU}}(\mathbf{x})\big) \end{bmatrix} \tag{7}$$

    where:
    - $W = R^{-1/2}$ is the individual 3×3 station weight matrix, which is the same for both of the two stations.
    - $\mathbf{I}_{N_{\text{T}}}$ and $\mathbf{I}_{N_{\text{K}}}$ are identity matrices of dimensions N_{T} and N_{K} (representing the number of observation epochs for TROLL and KOUROU, respectively).
    - $\otimes$ denotes the Kronecker product, which replicates the 3×3 matrix W along the diagonal to form the global weight matrices of dimensions $3N_{\text{T}} \times 3N_{\text{T}}$ and $3N_{\text{K}} \times 3N_{\text{K}}$.
    - $\mathbf{H}_{\text{stn}}(\mathbf{x})$ is the fully concatenated vector of predicted observations across all epochs for the respective station.
  - The `lsqnonlin` solver then iteratively minimizes $\|\mathbf{r}\|^2$ to find the optimal state and the associated *a posteriori* covariance.

- **Results & Analysis:** The state uncertainties are summarised in Table 11 and the estimated state vectors at t_i are reported in Table 12.

Table 11: State uncertainties at estimation epoch t_i.

Case	σ_r [km]	σ_v [km/s]	σ_a [km]
(v) TR+KO All (Kep)	1.697	0.0002	0.0689
(vi) TR+KO All (J_2)	1.832	0.0002	0.0763

Table 12: Navigation solution at t_i for multi-station cases.

Case	r_x [km]	r_y [km]	r_z [km]	v_x [km/s]	v_y [km/s]	v_z [km/s]
(v)	−27388.26	137.14	12200.47	1.99553	0.06699	−4.20242
(vi)	−27395.96	124.72	12188.27	1.99746	0.06838	−4.20130

Combining TROLL and KOUROU improves precision evidently: in case (vi), σ_r is 1.832 km, σ_v is 0.0002 km/s, and σ_a is 0.0763 km (vs. 40.0 km, 0.0047 km/s, and 47.4 km in case iv). This improvement stems from a longer data arc and superior geometry that better condition the estimation problem.

While both stations' measurements are generated from an SGP4 truth and the predicted measurements rely on Keplerian or J_2 dynamics, a propagation bias arises more evidently over the ≈ 13-hour KOUROU arc that exceeds the 0.01 km range uncertainty floor; as KOUROU dominates the least-squares cost by contributing 764 samples against TROLL's 69, the optimizer shifts the estimated state at t_i to absorb this accumulated bias over the full arc rather than minimizing the instantaneous error (note: this expected behavior does not invalidate the covariance reduction or navigation solution).

The choice of dynamics remains marginal, with σ_r varying only between 1.697 km (Keplerian) and 1.832 km (J_2). Final residuals for both stations are zero-centered and bounded within $\pm 3\sigma$ with no visible trends, confirming successful convergence.

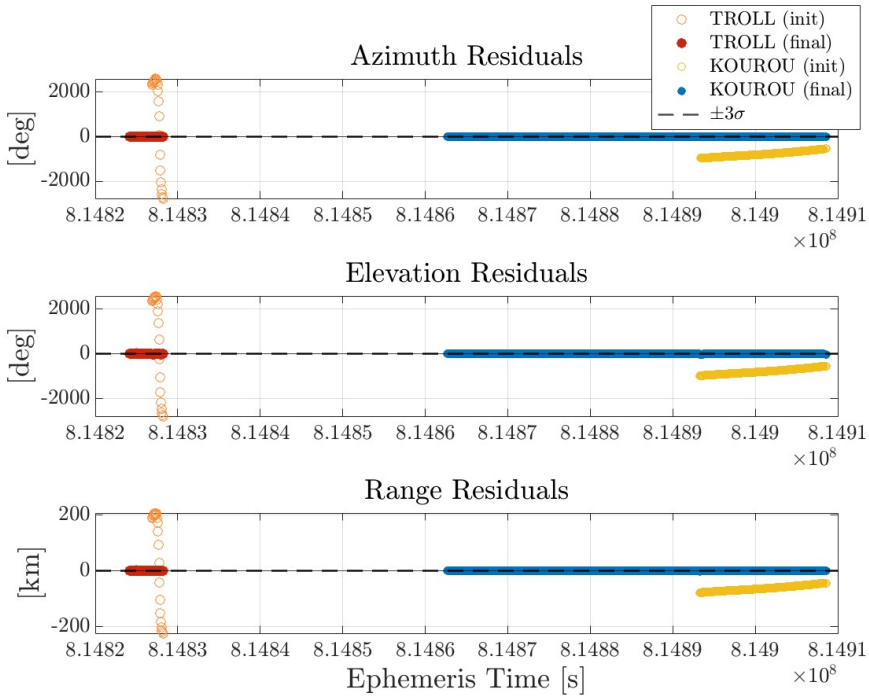


Figure 17: Unweighted measurement residuals case (vi): TROLL+KOUROU angles+range (J_2).

2.5 Performance Assessment

- **Objective:**

  To assess long-term performance of the six estimated states by propagating them over six orbital periods and comparing against SGP4 ground-truth.

- **Method:**

  – Each of the six optimal estimated states $\hat{\mathbf{x}}_k(t_i)$ (with $k = 1, \ldots, 6$) obtained at the estimation epoch t_i were propagated over six orbital periods using J_2 dynamics. The SGP4 ground-truth trajectory was similarly propagated over the same interval $[t_i, t_f]$. Despite different estimation models (Keplerian for cases i, iii, v, and J_2 for cases ii, iv, vi), all propagations use uniform J_2 dynamics to isolate the effect of the initial

state estimate. The J_2–SGP4 mismatch is systematic and common to all cases, so relative performance differences reflect estimation quality rather than propagation model choice.

– State error norms were computed at each time step

$$\|\mathbf{x}_k(t) - \mathbf{x}_{\text{SGP4}}(t)\| = \sqrt{(\mathbf{x}_k(t) - \mathbf{x}_{\text{SGP4}}(t))^T (\mathbf{x}_k(t) - \mathbf{x}_{\text{SGP4}}(t))}, \tag{8}$$

yielding the position and velocity error norms:

$$\|\Delta\mathbf{r}(t)\|_2 = \|\mathbf{r}_k(t) - \mathbf{r}_{\text{SGP4}}(t)\|_2, \quad \|\Delta\mathbf{v}(t)\|_2 = \|\mathbf{v}_k(t) - \mathbf{v}_{\text{SGP4}}(t)\|_2 \tag{9}$$

- **Results & Analysis:**

The error evolution over six orbital periods are shown in Figures 18 and 19 .

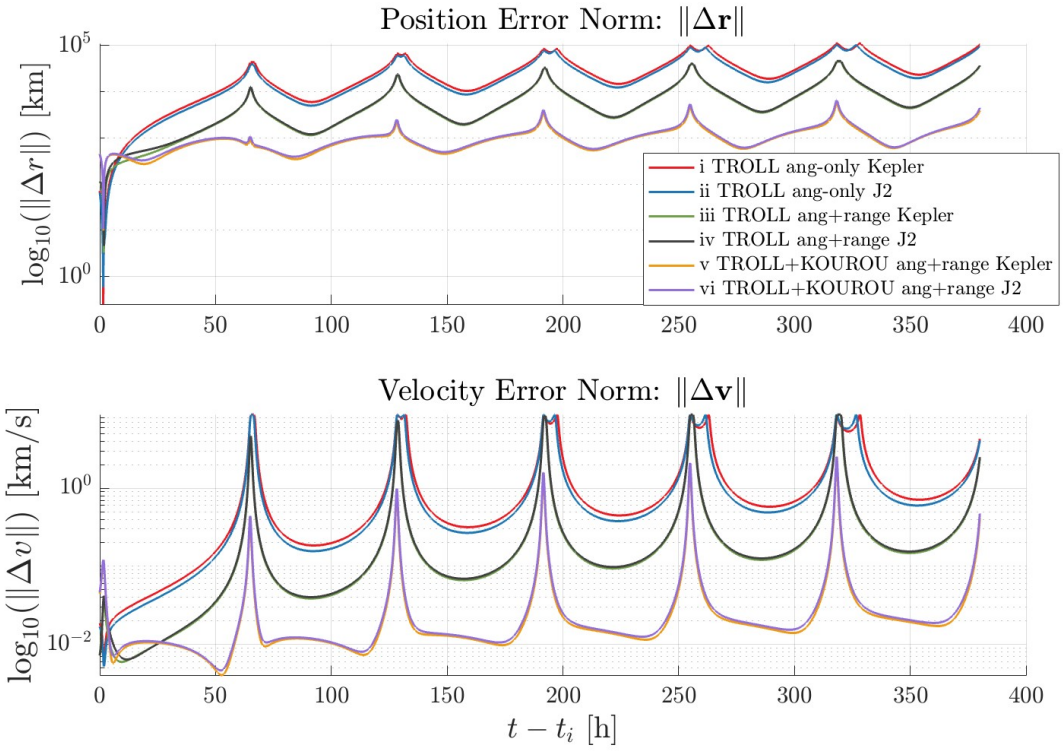


Figure 18: Position and velocity error norms over six orbital periods . The labels "Keplerian" and "J_2" refer to the dynamics used during the *estimation stage* at t_i, not during propagation.

Error growth exhibits periodic oscillations due to the high eccentricity trajectory, with error peaks coinciding with perigee passages where the orbital velocity is highest and the gravitational gradient is steepest; in this high-dynamic region, minor perturbations in the initial state are rapidly amplified into substantial trajectory divergences.

Multi-station cases (v, vi) achieve the best performance, with position errors remaining below $\sim 10^3$ km throughout the six-orbit horizon. Their superior accuracy reflects the complementary geometry of TROLL and KOUROU, which provides strong constraints on all orbital elements during estimation. Single-station cases show much larger divergence: angles-only solutions (i, ii) reach $\sim 10^5$ km, while the addition of range data (iii, iv) reduces this to $\sim 10^4$ km.

Performance improvements across the different considered cases stem primarily from adding range measurements and combining two stations rather than from the choice of

estimation dynamics, since the Keplerian vs. J_2 selection at the estimation stage is subordinate to the systematic J_2–SGP4 mismatch that governs long-term error growth, in fact SGP4 models additional perturbations beyond J_2, so any initial state differences at t_i are rapidly overwhelmed by accumulated propagation errors over the six-orbit arc regardless of which dynamics were used during estimation.

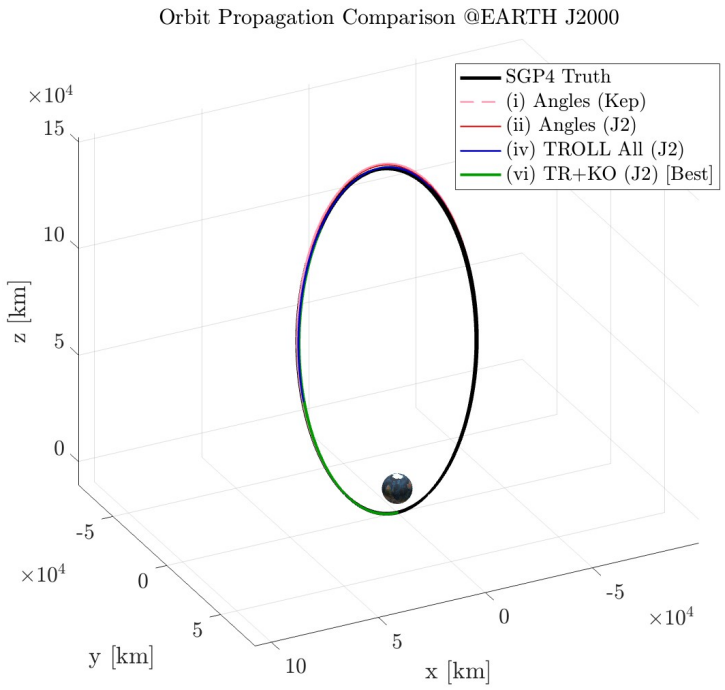


Figure 19: 3D orbit propagation in ECI J2000. SGP4 truth (black) and multi-station case (vi) (green) remain nearly coincident; angles-only case (i) (pink dashed) drifts significantly.

Exercise 3: Sequential filters

FUTURE (Fully autonomous feaTUre Recognition planetary Explorer) is an ASI CubeSat mission designed to advance spacecraft autonomy and reduce dependence on ground support operations. The mission is scheduled for launch in 2027 and will be released on a Sun-Synchronous Orbit at around 600 km altitude. The key objectives of the mission are:

- Develop autonomous position knowledge capabilities using AI algorithms to identify Earth surface features from space.
- Test advanced autonomous navigation technologies and AI Image Processing (IP) algorithms on-board.
- Validate position accuracy against GNSS data while operating in Low Earth Orbit (LEO).

The AI IP algorithm can identify the presence of Earth features in the images acquired by the on-board camera and, from these features, estimate the ECEF position expressed as latitude, longitude, and slant distance coordinates. Such measurements are only available within a given acquisition window, which is a time period during which the satellite is flying over coastlines, lakes, and other relevant ground features. During an acquisition window the sequential filter can perform both prediction and update steps, while outside of the acquisition window it can only perform a prediction.

You have been asked to perform an assessment of the on-board navigation filter performance, by comparing the on-board estimation from AI features to the one obtained using GNSS data. To perform the requested task you can refer to the following points.

1. *Check the acquisition windows*: Considering the initial state $\mathbf{x}_0$ and the time interval with a time-step of 5 seconds from t_0 to t_f reported in Table 13, predict the trajectory of the satellite in an inertial Earth-centered reference frame assuming Keplerian+J2 motion. Plot the ground-track and highlight the acquisition windows (*Hint*: simply plot in different color the part of the ground-track corresponding to the epochs of the acquisition windows provided in the file `ai_ip_measurements.csv`).

2. *Simulate GNSS measurement*: using the SPK kernel of the ground truth, simulate the GNSS measurements by adding a random error to the spacecraft position vector. Assume a Gaussian model to generate the random error, with noise provided in Table 13 for the components of the position vector. Verify (graphically) that the applied noise level is within the desired boundaries.

3. *Estimate the satellite state using GNSS measurements*: develop a sequential filter to narrow down the uncertainty on the knowledge of the satellite state vector. To this aim, you shall exploit the measurements computed at the previous point. Using an Unscented Kalman Filter (UKF), provide an estimate of the spacecraft state (in terms of mean and covariance) by sequentially processing the acquired measurements in chronological order. To initialize the filter in terms of initial state and covariance, use the initial state $\mathbf{x}_0 = \{\mathbf{r}_0, \mathbf{v}_0\}$ and initial covariance $\mathbf{P}_0$ reported in Table 13. We suggest using $\alpha = 0.01$ and $\beta = 2$ to tune the UT in this case.

4. *Estimate the satellite state using AI IP measurements*: develop a sequential filter using the AI IP measurements provided in the file `ai_ip_measurements.csv`. The measurements at each acquisition opportunity are the satellite radial distance, longitude, and latitude coordinates in the ECEF frame. We suggest using $\alpha = 0.01$ and $\beta = 2$ to tune the UT in this case. (*Hint*: when processing the longitude measurement, remember to consider the angle periodicity)

5. *Compare the two solutions*: for each estimated solution, plot the time evolution of the error estimate (computed with respect to the SPK kernel) together with the 3σ of the estimated covariance for both position and velocity. Compare the results and assess the expected on-board filter performance (inside and outside the acquisition windows).

3.1 Check the Acquisition Windows

- **Objective:** To predict the satellite's trajectory using a Keplerian+J_2 model, compute its ground track, and visualize the specific windows during which the AI IP system can acquire measurements.

- **Methodology:**

  The initial state $\mathbf{x}_0$ and time interval $[t_0, t_f]$ were taken from Table 13. The interval was discretized with $\Delta t = 5$ s generating `tspan` and yielding to $N = 7920$ samples over an 11 h arc. The nominal trajectory was propagated under Keplerian + J_2 dynamics. The ground track was retrieved via `compute_ground_track` function, which converts each ECI position to geodetic coordinates (longitude, latitude) through an ECI-to-ECEF rotation, with `wrapToPi` applied to ensure longitude continuity. To define the observation schedule, AI-IP measurement epochs from `ai_ip_measurements.csv` were synchronised with the simulation time grid using nearest-neighbour search (`dsearchn`). This mapping generated

Table 13: Initial conditions for FUTURE (ECI).

Parameter	Value
Initial state $\mathbf{x}_0$ [km, km/s]	$\mathbf{r}_0$ = [-436.489, 872.269, -6904.868] $\mathbf{v}_0$ = [7.126589, 2.598426, -0.131473]
Initial time t_0 [UTC]	2027-05-02T01:00:00
Final time t_f [UTC]	2027-05-02T12:00:00
Covariance $\mathbf{P}_0$ [km^2, km^2/s^2]	`diag([25,25,25,0.005,0.005,0.005])`
GPS Measurements noise	σ_p = 20 m
AI IP Measurement noise	$\sigma_{Lon,Lat}$ = 0.07 deg; σ_R = 10.12 km

a sequence of mapped indices, representing the specific integer steps of the simulation grid at which measurements are available. Discrete acquisition windows were delimited by identifying gaps in the mapped index sequence that exceeded a single time step.

- **Results & Analysis:** Figure 20 shows the 11-hour ground track, with 905 measurements mapped into 14 orange-highlighted acquisition windows with duration ranging from 1.08 min to 6.92 min. This underscores the primary challenge for the filter: maintaining state knowledge through the extensive J_2-perturbed propagation periods between these short, intermittent updates.

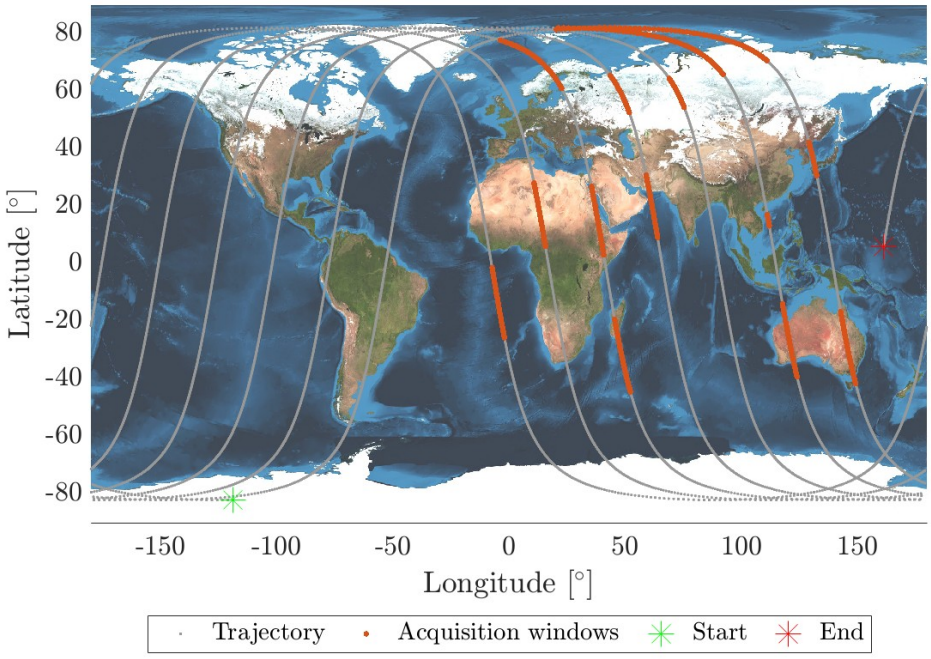


Figure 20: FUTURE Ground Track with Highlighted AI Acquisition Windows.

3.2 Simulate GNSS Measurement

- **Objective:** To generate a realistic, continuous stream of noisy GNSS position data and to graphically verify that the applied noise is within the desired statistical boundaries.

- **Methodology & Code Implementation:**

  The ground truth was retrieved from SPK kernel at each epoch in `tspan` which retrieves the full spacecraft state in the J2000 frame. Per Table 13, the noise on each ECI axis is independent and identically distributed (i.i.d.) Gaussian with $\sigma_p = 20$ m, giving a diagonal covariance:

$$R_{\text{GNSS}} = \sigma_p^2 \, I_{3\times 3}$$

  Synthetic measurements were then generated as:

$$\mathbf{r}_{\text{GNSS}}(t_k) = \mathbf{r}_{\text{true}}(t_k) + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0},\, R_{\text{GNSS}})$$

  with a fixed random seed for reproducibility. Noise consistency was verified by plotting the absolute error $|\mathbf{r}_{\text{true},i}(t_k) - \mathbf{r}_{\text{GNSS},i}(t_k)|$ per ECI component against the theoretical $+3\sigma$ bound, confirming zero-mean behaviour with no visible trend or bias.

- **Results & Analysis:** As shown in Figure 21, the injected noise is effectively zero-mean (empirical means < 0.3 m), with empirical standard deviations of 19.80 m, 20.12 m, and 20.09 m closely matching the target $\sigma_p = 20$ m. The fraction of samples within 3σ exceeds 99.7 % on all axes, with no visible trend or bias over the 11 h arc, confirming the validity of the stochastic model.

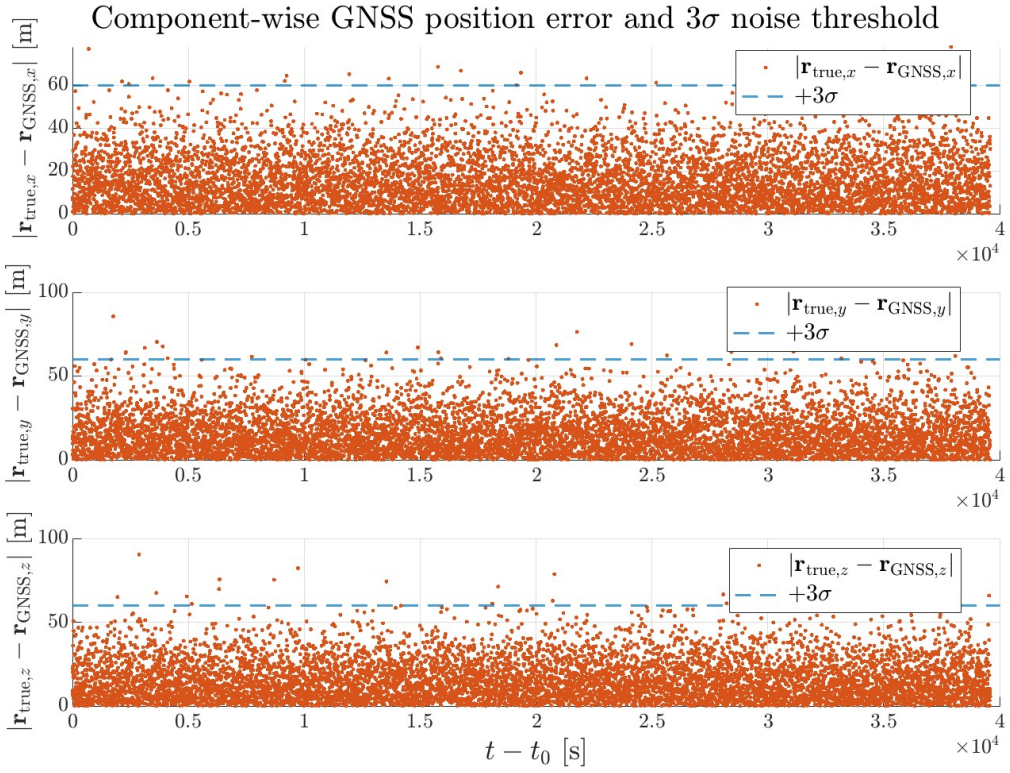


Figure 21: Verification of Simulated GNSS Noise with 3σ Bounds.

3.3 State Estimation with UKF (GNSS and AI IP)

- **Objective:** To develop and run two separate Unscented Kalman Filters: one processing the continuous GNSS data (Point 3) and another processing the sparse AI IP data (Point 4).

- **Theoretical Framework: The Unscented Kalman Filter (UKF)**

  To estimate the satellite state, an Unscented Kalman Filter (UKF) was implemented. The Unscented Kalman Filter (UKF) is a recursive estimator for nonlinear systems. Rather than linearizing the dynamics, it propagates a set of deterministically chosen sample points (sigma points) through the true nonlinear functions via the Unscented Transform (UT), reconstructing the state mean and covariance from the results. The algorithm begins with an initial estimate of the state and its uncertainty and then proceeds in a step-by-step recursive loop.

  1. **Initialization**

     The filter is initialized with the a priori state estimate and its corresponding covariance at the initial time t_0.

     – Initial state: $\hat{x}_0 = E[x_0]$
     – Initial covariance: $P_0 = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T]$

  2. **Set Unscented Transform Parameters**

     The behavior of the UT is governed by a set of scaling parameters:

     – α: Controls the spread of the sigma points (typically a small value, e.g., 0.01).
     – β: Incorporates prior knowledge of the state's distribution ($\beta = 2$ is optimal for Gaussian distributions).
     – κ: A secondary scaling parameter, often set to 0.

     These are used to compute a composite scaling parameter $\lambda = \alpha^2(n + \kappa) - n$, where n is the dimension of the state vector ($n = 6$). The weights for reconstructing the mean (W_m) and covariance (W_c) from the sigma points are then computed:

$$W_m^{(0)} = \frac{\lambda}{n+\lambda}, \quad W_c^{(0)} = \frac{\lambda}{n+\lambda} + (1 - \alpha^2 + \beta), \quad W_m^{(i)} = W_c^{(i)} = \frac{1}{2(n+\lambda)} \quad \text{for } i = 1, \ldots, 2n \tag{10}$$

  3. **The Recursive Prediction-Update Loop**

     For each time step k, the filter performs the following two stages:

     – **Prediction (Time Update):**
     This stage propagates the state estimate and covariance from t_{k-1} to t_k using the nonlinear system dynamics $f(x, t)$. Process noise is neglected ($Q = 0$), as the Keplerian $+ J_2$ model is assumed sufficiently accurate.

       * Generate Sigma Points: $2n+1$ sigma points are generated from the previous a posteriori state $\hat{x}_{k-1}$ and covariance P_{k-1}:

$$\begin{cases} \chi_{k-1}^{(0)} = \hat{x}_{k-1} \\ \chi_{k-1}^{(i)} = \hat{x}_{k-1} + \left[\sqrt{(n+\lambda)P_{k-1}}\right]_i \\ \chi_{k-1}^{(i+n)} = \hat{x}_{k-1} - \left[\sqrt{(n+\lambda)P_{k-1}}\right]_i \end{cases}, \quad i = 1, \ldots, n \tag{11}$$

       * Propagate Sigma Points: Each sigma point is propagated through the nonlinear Keplerian $+ J_2$ dynamics:

$$\chi_{k|k-1}^{(i)} = f\left(\chi_{k-1}^{(i)}\right) \implies \begin{cases} \hat{x}_{k|k-1} = \sum_{i=0}^{2n} W_m^{(i)} \chi_{k|k-1}^{(i)} & \text{(predicted mean)} \\ P_{k|k-1} = \sum_{i=0}^{2n} W_c^{(i)} \Delta\chi^{(i)} \left(\Delta\chi^{(i)}\right)^T & \text{(predicted covariance)} \end{cases} \tag{12}$$

where $\Delta\chi = \chi_{k|k-1}^{(i)} - \hat{x}_{k|k-1}$.

- **Update (Measurement Update):**
  This stage corrects the a priori estimate using a new measurement z_k.
  - Project sigma points to measurement space and predict mean and covariance:

$$z_k^{(i)} = h\left(\chi_{k|k-1}^{(i)}\right), \quad \hat{z}_k = \sum_{i=0}^{2n} W_m^{(i)} z_k^{(i)}, \quad S_k = \sum_{i=0}^{2n} W_c^{(i)} \left[z_k^{(i)} - \hat{z}_k\right] \left[z_k^{(i)} - \hat{z}_k\right]^T + R$$

    where R is the measurement noise covariance matrix. For GNSS tracking, it is isotropic ($R = \sigma_p^2 I_{3\times 3}$ with $\sigma_p = 20\,\text{m}$), while for AI-IP it is a diagonal matrix: $R = \text{diag}(\sigma_\rho^2, \sigma_\lambda^2, \sigma_\phi^2)$ with $\sigma_\rho = 10.12\,\text{km}$ and $\sigma_{\lambda,\phi} = 0.07°$.
  - Compute cross-covariance and Kalman gain:

$$P_{xz,k} = \sum_{i=0}^{2n} W_c^{(i)} \left[\chi_{k|k-1}^{(i)} - \hat{x}_{k|k-1}\right] \left[z_k^{(i)} - \hat{z}_k\right]^T, \quad K_k = P_{xz,k} S_k^{-1}$$

  - Update state and covariance:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \left(z_k - \hat{z}_k\right), \quad P_{k|k} = P_{k|k-1} - K_k S_k K_k^T$$

  This process is repeated at each time step, recursively updating the state estimate and its uncertainty. *Note:* throughout, z_k denotes the actual measurement, $z_k^{(i)}$ the predicted measurement for the i-th sigma point, and $\hat{z}_k$ the predicted measurement mean.

- **Implementation & Code Logic**

  Two separate helper functions, `UKF_GNSS` and `UKF_AIIP`, were implemented to encapsulate this logic.

  - **UKF for GNSS (Point 3):**
    - This function implements the standard UKF cycle described above. The filter runs continuously, performing both a prediction and an update at every 5-second time step.
    - **Measurement Model $h(\mathbf{x})$:** Since both the filter state and GNSS measurements are in the ECI frame, the predicted measurement $z_k^{(i)} = h(\chi_{k|k-1}^{(i)})$ is a simple extraction of the first three position components of the propagated sigma point $\chi_{k|k-1}^{(i)}$, and the actual measurement $z_k = \mathbf{r}_{\text{GNSS}}(t_k)$ is the noisy ECI position from the simulated GNSS data.
  - **UKF for AI-IP (Point 4):**
    - **Sparse Updates:** At each epoch, the logical array `ai_visible` flags measurement availability. Inside acquisition windows, the full prediction-update cycle is executed; outside, the update is skipped and the filter performs pure prediction, with the posterior reducing to the prior: $\mathbf{x}_k^+ = \mathbf{x}_k^-$, $P_k^+ = P_k^-$.

– **Measurement Model $h(\mathbf{x})$:** For each sigma point, the predicted measurement is obtained by rotating the ECI position to ECEF (`cspice_pxform`) and converting to spherical coordinates $[\rho, \lambda, \phi]^T$ via `cspice_reclat`, where ρ is the geocentric radius and (λ, ϕ) are longitude and latitude in the Earth-fixed frame. The actual measurement $z_k = [\rho_k,\, \lambda_k,\, \phi_k]^T$ is read directly from the `csv` file. Longitude periodicity is handled by wrapping to $[-\pi, \pi]$ via `wrapToPi`; the predicted longitude mean is computed via circular averaging, residuals via `angdiff`, while latitude is treated as non-periodic.

3.4 Compare the Two Solutions

- **Objective:** To assess and compare the performance of the two UKF implementations (GNSS-based and AI-IP-based) by analyzing the time evolution of estimation errors and formal 3σ uncertainty bounds, with particular attention to behavior inside and outside the AI-IP acquisition windows.
- **Methodology:**
  The reference 6D truth was retrieved by querying the SPK kernels . For each filter, position and velocity error norms were computed as:

$$e_r(t_k) = \|\hat{\mathbf{r}}(t_k) - \mathbf{r}_{\text{true}}(t_k)\|, \quad e_v(t_k) = \|\hat{\mathbf{v}}(t_k) - \mathbf{v}_{\text{true}}(t_k)\|$$

  The component-wise 3σ bounds are extracted from the diagonal of the 6×6 covariance:

$$\sigma_{3,r,i}(t_k) = 3\sqrt{P_{r,ii}(t_k)}, \quad \sigma_{3,v,i}(t_k) = 3\sqrt{P_{v,ii}(t_k)}, \quad i = 1, 2, 3$$

  and combined into a scalar bound $\sigma_{3,r} = \sqrt{\sum_i \sigma_{3,r,i}^2}$ (and analogously $\sigma_{3,v}$) for direct comparison against the error norm. Time-series plots of e_r, e_v overlaid with their 3σ bounds are shown in Figures 22 and 23 .
- **Results & Analysis:**
  – **GNSS Filter Performance:** Figures 22 and 23 shows the filter error evolution over the full 11 h arc. After an initial transient driven by the mismatch between the a priori state estimate and the truth (rapidly corrected by successive GNSS updates) the solution converges to a steady-state RMS position error of 0.020 km, in line with the $\sigma_p = 20$ m sensor noise floor. The final position error reaches 0.0038 km, with RMS velocity error of 6.79×10^{-4} km/s and final velocity error of 3.73×10^{-6} km/s . The error norm remains strictly below the 3σ bound at all times, confirming that the filter is statistically consistent. Note that the specific numerical results depend on the chosen random seed.
  – **AI-IP Filter Performance:** As visible in Figures 22 and 23 , the AI-IP error trace exhibits a distinctive sawtooth pattern: uncertainty accumulates during coast arcs where only orbital propagation is active, and drops sharply at each of the 905 acquisition epochs distributed across the 11 h arc. The RMS position error settles at 3.31 km with a final position error of 0.705 km, while the velocity RMS is 5.98×10^{-3} km/s with a final velocity error value of 5.46×10^{-4} km/s, reflecting the combined effect of sparse updates and the comparatively large geodetic measurement noise. Throughout the full arc, the error norm remains strictly below the 3σ bound, confirming statistical consistency and that the filter never becomes overconfident despite the intermittent measurement availability.
  – **Comparison:** The two filter configurations are compared in terms of global RMS errors and window-segmented statistics to isolate the effect of measurement availability and quality.

The AI-IP solution is penalised by higher-noise geodetic measurements and sparse update availability, with uncertainty growing between consecutive acquisitions as detailed above; a window-segmented breakdown further quantifies this effect showing that position RMS drops from 3.49 km outside to 1.17 km inside the windows, with a consistent trend for velocity (6.32×10^{-3} vs. 1.77×10^{-3} km/s). By contrast, evaluating the GNSS error over the same AI-IP window intervals (for reference only) yields position RMS of 0.0074 km inside and 0.0216 km outside, values that remain uniformly small in both regimes, underscoring how large the performance gap is with respect to the AI-IP filter regardless of whether an acquisition window is active or not.
The two order of magnitude accuracy gap confirms that the AI-IP system, while viable as an independent backup during GNSS outages,cannot match the navigation precision achievable with continuous high-accuracy position measurements.

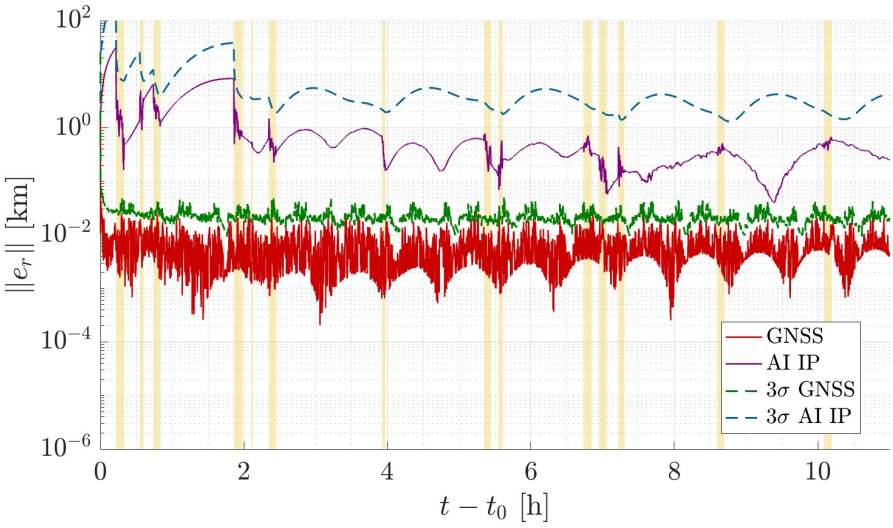


Figure 22: Time evolution of position estimation error norms with corresponding 3σ bounds for GNSS and AI-IP filters. Shaded regions indicate AI-IP acquisition windows.

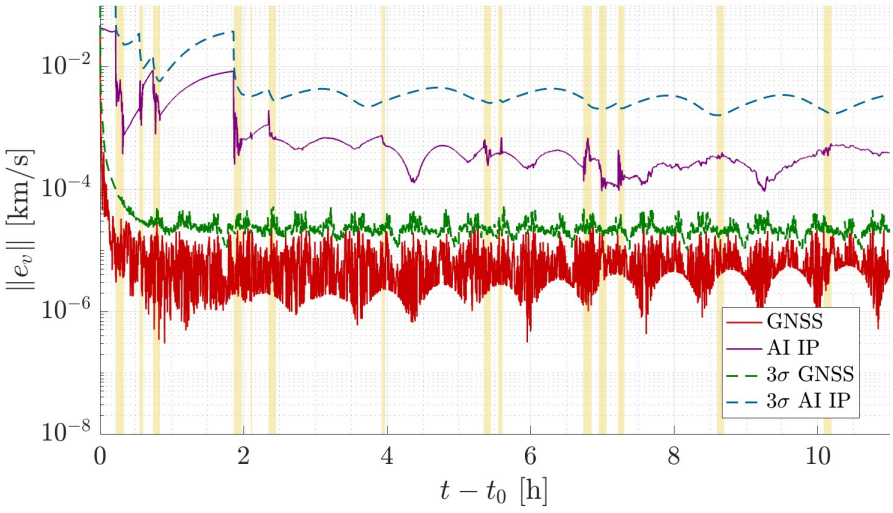


Figure 23: Time evolution of velocity estimation error norms with corresponding 3σ bounds for GNSS and AI-IP filters. Shaded regions indicate AI-IP acquisition windows.